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Multiplanar Deep Learning for Hepatic Vessel Segmentation: A Systematic Ablation Study

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I declare that this dissertation is all my own work, except as indicated in the text.

At a Glance (TL;DR)

This page provides a one-page summary of the entire dissertation for quick orientation.

	Key Point
Problem	Hepatic vessel segmentation from CT is hard: vessels occupy less than 3% of liver volume, peripheral branches taper to sub-millimetre widths, and single-orientation (axial) models miss branches that are better resolved in other anatomical planes
Method	Train three independent nnU-Net v2 3D_fullres models on axial, sagittal, and coronal reorientations of MSD Task08. Combine predictions with majority voting (2-of-3 agreement per voxel). Also evaluate: clDice topology-preserving loss, learned CNN fusion, and connected-component post-processing.
Key Results	Multiplanar majority voting improves average Dice from 59.65% → 64.93% (+5.29%, $p = 0.0045$). Tumour Dice gains +10.63% ($p = 0.0029$). Combining multiplanar fusion with clDice loss achieves the best topology score (clDice = 70.47%) and the highest tumour Dice among non-post-processed methods (66.34%); the post-processed clDice variant reaches 66.60% tumour Dice but at the cost of vessel Dice.
Negative Results	(1) Connected-component post-processing hurts vessel Dice — genuine peripheral branches are removed. (2) Training on all orientations pooled into one dataset fails completely (vessel Dice = 0%). (3) Learned CNN fusion underperforms simple voting with limited training data.
Takeaway	Multiplanar fusion is a simple, effective plug-in improvement for nnU-Net vessel pipelines. Train one model per orientation, then vote. clDice adds complementary topology benefits. Post-processing requires careful calibration to avoid damaging genuine vessel anatomy.

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Abstract

Hepatic vessel segmentation from computed tomography (CT) scans is vital for surgical planning, tumour staging and liver transplant screening. But the tree-like topology, high class imbalance (vessels constitute less than 3% of the liver volume) and poor contrast of small branches pose a significant challenge for automatic hepatic vessel segmentation. This thesis explores a multiplanar deep learning strategy with the nnU-Net architecture to enhance hepatic vessel segmentation in the Medical Segmentation Decathlon Task08 dataset.

We trained three nnU-Net v2 models on different orientations of the same dataset (axial, sagittal and coronal), which learned distinct features for each orientation. These models' predictions were combined through majority voting at the voxel level and showed an improvement of the average Dice score from 59.65% (axial baseline) to 64.93% on an internal validation split of 20 cases, a statistically significant difference ($p = 0.0045$, Wilcoxon signed-rank test). The tumour segmentations were most improved from 53.26% to 63.89% ($p = 0.0029$). These results are evaluated on this split and cannot be directly compared to published metrics on the official 140-case MSD test split. Various fusion methods such as average voting, weighted voting and a learned CNN-based fusion were also tested; all voting-based fusion methods were equivalent, whereas the CNN-based fusion was inferior to majority voting under the data-limited constraints of this project.

Further, the centreline Dice (clDice) topology-aware loss was incorporated into training via a custom nnU-Net trainer with a warmup-then-ramp learning rate schedule. nnU-Net's deep supervision approach caused training divergence with standard clDice weights ($\alpha = 0.5$); stable convergence was achieved by lowering the weight to 0.1 with a 50-epoch warmup - a practical consideration not mentioned in previous clDice studies and relevant to any topology-preserving nnU-Net-based pipeline. Although training only on axial clDice was not entirely successful - boosting topology sensitivity while sacrificing volumetric Dice - multiplanar fusion of all three clDice-trained models outperformed all non-post-processed experiments in terms of topology (clDice = 70.47%) and tumour Dice (66.34%). Connected-component post-processing was explored but found to reduce vessel Dice by eliminating small vessel fragments and constitutes a negative result.

In summary, this research demonstrates that multiplanar fusion is a simple but effective strategy for enhancing hepatic vessel segmentation when using the nnU-Net framework, and that multiplanar fusion and topology-aware training combine well for improved vessel segmentation and tumour segmentation.

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Chapter 1. Introduction

1.1 Clinical Context and Motivation

Hepatic vessels are key to surgical planning. The portal venous system, which brings nutrient-laden blood from the gut; and hepatic venous system, which drains blood to the inferior vena cava, divide the liver into eight functionally distinct Couinaud segments [7, 26]. Precise delineation of these vessels from computed tomography (CT) images is essential for successful liver resections, transplant donor assessment and staging of liver tumours. If the surgeon can accurately predict the vessel distribution before surgery, they can determine the extent of resection required to maintain sufficient blood flow to the remaining liver parenchyma, thus minimising the risk of post-operative liver failure.

Despite its clinical utility, liver vessel segmentation remains a challenging task in abdominal CT imaging. Hepatic vessels have a tree-like structure with a wide range of diameter: portal trunks are 10-15 mm wide, but small branches are as small as sub-millimetre diameter, at the limit of clinical CT resolution [1, 6]. This results in class imbalance, with vessel voxels accounting for less than 3% of the liver volume. Lumen-to-liver contrast for small branches can be poor and is further eroded by noise and partial volume effects. Furthermore, there is significant inter-person variability in vessel branching patterns, making shape priors (that are effective in other organs) less effective.

1.2 Problem Statement

Existing deep learning methods for hepatic vessel segmentation typically operate on volumes from a single orientation - the axial plane as acquired by the CT scanner. Although state-of-the-art methods such as nnU-Net [15, 16] have reported excellent results through the ability to automatically configure the training pipeline for each dataset, models trained on single anatomical orientations can overlook features that are better captured in alternative orientations. A small branch of a vessel that is oriented parallel to the axial plane, for instance, may only appear on one or two slices in the axial plane, but can be seen on dozens of slices in the sagittal and coronal planes. This insight suggests the focus of this study: is it possible to train multiple models on different anatomical orientations and merge their predictions to improve hepatic vessel segmentation?

Another issue is the lack of correspondence between traditional segmentation measures and clinically important vessel connectivity. The Dice coefficient is a voxel-based metric, so a segmentation can score highly in Dice without preserving vessel connectivity: a good Dice score can be achieved with a topologically disconnected vessel tree. Topologically sensitive loss functions such as the centreline Dice (clDice) [20] penalise connectivity issues during training but have not been explored in conjunction with multiplanar fusion for hepatic vessels.

1.3 Approach Overview

This project trains three separate nnU-Net v2 models (3D_fullres) on the Medical Segmentation Decathlon Task08 Hepatic Vessel dataset [40, 41]. The dataset is rotated to produce sagittal and coronal volumes; each model is trained on volumes in its respective orientation. During evaluation, the three models' predictions are re-oriented to the axial orientation and then fused via majority voting - a rule that assigns a label to a voxel if two or more models agree.

The pipeline is augmented with two additional components: (1) a custom nnU-Net trainer with the clDice loss function and warm-up then ramp-up schedule for training-time topology regularisation, and (2) post-processing using connected-component filtering. Various fusion approaches are explored, including simple average and weighted voting, and learned CNN-based fusion. The impact of each component is assessed in an ablation study.

1.4 Research Questions and Hypotheses

The aim of this dissertation is to answer the following questions:

RQ1: Is accuracy improved by fusing predictions from axial, sagittal and coronal nnU-Net models by majority voting, compared to a baseline single axial model?

RQ2: Does training with the clDice loss, which preserves vessel topology, improve vessel connectivity, and is this improvement preserved via multiplanar fusion?

RQ3: How do different fusion approaches (majority, average, weighted, learned) perform?

We expect that (H1) multiplanar fusion will improve Dice by averaging out orientation biases, (H2) training with clDice will improve topology metrics at the expense of Dice, and (H3) majority voting will outperform more complex fusion strategies given the small size of our validation set.

1.5 Scope and Evaluation Limitations

All scores in this dissertation are evaluated on the 20-case internal hold-out test set from the 303 labelled MSD Task08 cases, rather than the official 140-case MSD challenge test set. Since labels for this test set are not publicly available without submitting to the challenge server, we cannot directly compare our absolute Dice scores with those published, such as the official nnU-Net submission (81.64% vessel Dice) or Swin UNETR (70.03% average Dice), which are computed on the larger and more representative official test set. All experiments in this study are thus relative to improvements over our own axial baseline and are statistically significant according to the Wilcoxon signed-rank test over the same 20 cases. We further discuss this in Section 5.6.

Further, all our models are trained on fold 0 only, while the official nnU-Net submission is trained with 5-fold cross-validation and ensembling. This also contributes to the absolute score differences between our work and those reported in the literature. The focus of this work is the relative improvement in the multiplanar model, and the ablation evidence provided and not the absolute comparison to fully trained multi-fold networks.

Importantly, relative comparisons between methods are unaffected by the single-fold approach. All models in this work - the axial baseline and all ablations - are trained on the same data split (fold 0), with the same preprocessing steps, and tested on the same 20 test cases held out from fold 0. The choice of fold, then, is a constant variable to all experimental conditions. Any potential bias in fold 0 would be present in all models and cannot account for differences in improvement. This is standard practice in ablation studies in machine learning, where relative comparisons are the main evidence, and absolute results are only used to situate findings relative to external baselines. The time and resource constraints of the project on Google Colab Pro (12-14 hours of GPU training time per model, meaning a multiplanar 5-fold cross-validation would require 15 models and over 180 hours of training) precluded the use of 5-fold cross-validation in an undergraduate project. Multi-fold training is the most obvious next step (Section 6.3).

1.6 Contributions

This work makes the following contributions:

1. The multiplanar nnU-Net for hepatic vessel segmentation, which showed a statistically significant +5.29% boost in average Dice over the single-orientation baseline, thanks to majority voting fusion.

2. Thorough testing the integration of clDice loss into nnU-Net with a custom trainer using a warmup schedule, and observation of both the improvement in topology and training instability with deep supervision.

3. A full ablation study of four fusion strategies (majority, average, weighted voting, and learned CNN fusion - with the three voting variants confirmed empirically equivalent), three loss setups and post-processing, with statistical significance testing on 20 validation cases.

4. Negative results - multiplanar nnU-Net training by combining all orientations into a single dataset and connected-component post-processing have detrimental effects on vessel Dice scores - to inform future researchers.

1.7 Dissertation Structure

The remainder of this dissertation is organised as follows. Chapter 2 provides an overview of related literature in five topics: hepatic vessel anatomy, deep learning segmentation networks, multiplanar models, topological losses, and ensemble approaches. Chapter 3 outlines the method in detail, including preparation of the dataset, training of models, fusion strategies and evaluation procedures. Chapter 4 reports results of all ablation experiments. Chapter 5 examines the results, their significance and limitations. Finally, chapter 6 provides a conclusion and future directions.

Chapter 2. Literature Review

This chapter reviews the research areas on which this project is based. It starts with the medical need for hepatic vessel segmentation and the challenges that make it a complex task (Section 2.1). Then the basic deep learning architectures adopted for medical image segmentation are presented, focusing on the nnU-Net architecture adopted for our pipeline (Section 2.2). Multiplanar and multi-view methods - the key source of inspiration for this project - are discussed in Section 2.3, and topology-preserving loss functions (the centreline Dice (clDice) loss function we use for training) in Section 2.4. State-of-the-art techniques for hepatic vessel segmentation are reviewed in Section 2.5 and ensemble and fusion approaches in Section 2.6. The chapter concludes with an overview of the research gap (Section 2.7).

2.1 Clinical Context: Hepatic Vessel Anatomy and Segmentation

The liver receives blood from two main venous systems: the portal vein, bringing nutrient-laden blood from the gut, and the hepatic veins, which drain to the inferior vena cava [26]. These vascular trees segment the liver into eight functional, independent Couinaud segments - a classification that's key to surgical planning for liver resection, transplant donor assessment, and for cancer staging [26]. Segments are defined by segmenting these structures from CT scans to ensure safe surgery.

Vessel segmentation is a challenging task in abdominal CT images. Vessels have a tree-like structure with a wide range of diameters - portal trunks can be 10-15 mm in diameter, but branches narrow to less than 1 mm (close to the resolution of clinical CT) [1, 6]. This leads to a highly imbalanced classification problem: vessel voxels constitute less than 3% of the liver volume [36, 40]. Vessel lumen and parenchyma have low contrast, especially for small vessels, and are confounded by noise and partial volume effects [1, 2]. Vessel branching patterns are also highly variable from person to person, making it difficult to use prior anatomical information [3, 4].

The Medical Segmentation Decathlon (MSD) Task08 - Hepatic Vessel [40, 41] offers 443 portal-venous phase CT scans with voxel-level labels for hepatic vessels and tumours to provide a large public benchmark for this task, used in this project.

2.2 Deep Learning for Medical Image Segmentation

2.2.1 Foundational Architectures

Ronneberger et al.'s U-Net [28] pioneered the encoder-decoder with skip connections architecture that is the most common design choice for medical image segmentation; Çiçek et al. [29] subsequently adapted this to 3D volumes in the 3D U-Net, showing that dense segmentations can be learned from sparse labels - an architecture that underpins nnU-Net. Milletari et al. [30] proposed V-Net and the Dice-based loss function that is now ubiquitous for dealing with class imbalance in segmentation, which directly optimises voxel overlap without the need for class weighting. We adopt the default nnU-Net loss of a Dice and cross-entropy

(Dice+CE) combination for our baseline model. Later modifications like Attention U-Net [31], which leverages attention gates to filter out background and emphasise small structures, are also suitable for the thin structures of the hepatic vasculature.

2.2.2 The nnU-Net Framework

The automatic configuration of nnU-Net [15, 16] is probably the most significant advance in practical medical image segmentation in recent years. Instead of a new network architecture, nnU-Net showed that automation of the decisions around image pre-processing, augmentation, training, inference, and post-processing is more important than changes to the network architecture itself. It automatically selects the best resampling resolution, normalisation, patch size, batch size and architecture for a new dataset from a dataset fingerprint instead of requiring manual hyperparameter tuning that consumes weeks of expert time.

For our project, nnU-Net v2.6.2 chooses the 3D_fullres configuration for MSD Task08, which corresponds to a PlainConvUNet with six stages, (32, 64, 128, 256, 320, 320) feature maps, instance normalisation and LeakyReLU activations. The patch size is (64, 192, 192) with a batch size of 2, and the target spacing is (1.5, 0.799, 0.799) mm. We train all models on fold 0, which offers a 242/41 training/validation split from the 283 cases available after hold-out, with 20 held-out cases left for testing.

A recent large-scale evaluation [17] confirmed the framework's current relevance, with appropriately tuned CNN-based U-Net variants still performing as well as, or better than, transformer-based and Mamba-based networks in controlled experiments. The study found many transformer claims were due to poor baselines or insufficient computational considerations.

2.2.3 Transformer and Alternative Architectures

Despite Isensee et al. [17], transformers have sparked considerable interest. TransUNet [32] uses CNN features as input to a Transformer for long-range dependency modelling; UNETR [33] uses a pure Vision Transformer encoder with a CNN decoder; Swin UNETR [34, 35] uses a hierarchical window-based transformer (Swin transformer) instead of a standard ViT. Within the nnU-Net framework, the nnFormer [18] alternates convolution and volume-based self-attention, and MedNeXt [19] achieves the best results on four MSD tasks using a ConvNeXt-like backbone.

We use a vanilla CNN-based nnU-Net backbone, as per [17]. A caveat: that benchmarking study was performed by the authors of nnU-Net, which could be a source of bias; and independent verification is limited. In practice, CNN-based methods are easier to train, require less memory, and have better reproducibility - important considerations for a project using Google Colab with time constraints.

2.3 Multiplanar and Multi-View Approaches

This section reviews the methodological family most directly relevant to our project: approaches that combine information from multiple anatomical planes to improve 3D segmentation.

2.3.1 Origins and Formative Evidence

The notion of fusing multiple views to segment 3D structures dates back before deep learning. Roth et al. [8] showed that combining probability maps of axial, coronal and sagittal patch-based ConvNets helped segment pancreas in abdominal CT scans, concluding that different anatomical planes provide complementary spatial information. Although the patch-based approach has been replaced by fully convolutional networks, the use of view complementarity underpins the multiplanar approach.

This was formalised in the multi-planar CNN model of Mortazi et al. [9], where the authors trained three 2D CNNs (one per orientation) for cardiac substructure segmentation and fused their predictions with an adaptive fusion approach. They demonstrated that training on the much larger number of 2D slices was more practical than training a single 3D model on the small amount of volumetric data; and that multi-planar fusion restored the 3D context lost by the 2D models. This reasoning extends to our problem: while we train full 3D nnU-Net models per orientation rather than 2D CNNs, each orientation still captures distinct information about the vessel anatomy that can be leveraged by fusion.

Meyer et al. [14] demonstrated this for prostate segmentation from multi-planar MRI, where the addition of sagittal and coronal streams to an axial model improved segmentation; and that the improvement was in regions that are difficult to segment anatomically, due to low out-of-plane resolution in axial scans (equivalent to thin hepatic vessel branches that are poorly resolved in one view).

2.3.2 Multi-Planar U-Net and Systematic Comparisons

Perslev et al. [10] proposed the Multi-Planar U-Net (MPUnet), which samples 2D slices across different planes of a 3D volume and combines predictions across planes. Rather than just the axial, sagittal and coronal planes, MPUnet samples from arbitrary planes during training, effectively performing 3D data augmentation. This system achieved 5th and 6th ranks in the two phases of the 2018 Medical Segmentation Decathlon and performed well across all 10 tasks without task-specific training. The key insight from this work is that this test-time fusion results in a reduction of variance - as long as errors are independent across different views for a given subject. This is an important assumption for majority voting to work, and it seems to be true in practice due to different views making different mistakes.

The strongest empirical evidence for our approach comes from Zhang et al. [11], who performed a large-scale benchmark study of 2.5D segmentation methods on three typical medical segmentation tasks. They compared majority voting, weighted voting, and a learnable Volumetric Fusion-Net (VFN) that fused segmentations from the axial, sagittal and coronal views produced by 2D U-Nets. They found that: (1) all multi-view fusion methods were better than single-view 2D models; (2) multi-view fusion produced segmentation results on par with full 3D CNNs; and (3) for anisotropic volumes (e.g. where the z-axis is much lower resolution than the x- and y-axes), 3D CNNs sometimes underperform 2.5D methods because the low-resolution axis introduced extra features.

A critical point to note is that most of these papers use 2D models per view, not 3D models processing reoriented 3D volumes, as in our case. Each single model in a 3D nnU-Net learns information along all three axes via the 3D receptive field of its convolutional layers, but the difference between our orientation-specific models is which axis is learned with the higher-resolution in-plane patch size. At the dataset's anisotropic voxel spacing (1.5, 0.799, 0.799) mm, it determines which thin structures are contained within the high-resolution patch dimensions, and are thus learned best - the key to complementary information in the multiplanar fusion of a fully-3D approach.

2.3.3 Alternative Multi-View Formulations

It is not necessary to train multiple models in a multi-view approach. Yang et al. [13] introduced ACS convolutions, which divide 2D convolutional kernels into channel subsets and convolve each subset along a different anatomical axis in a single model, filling in the gap between 2D and 3D representation learning while maintaining compatibility with 2D pretrained weights. Our three independent nnU-Net models are not as pretty but more flexible - individual views can be retrained or replaced without affecting the others, and no kernel hacking is needed. Xia et al. [12] proposed uncertainty-aware multi-view co-training (UMCT) to use Bayesian deep learning to weigh each view's contribution by its prediction confidence during label fusion - which essentially incorporates prediction confidence into the voting step, rather than assuming all orientation models are equally good.

2.4 Topology-Preserving Loss Functions

Traditional segmentation losses such as Dice and cross-entropy optimise voxel overlap, assuming that each voxel is independent. This is problematic for tubular structures like blood vessels. An isolated mis-segmented voxel at a branch point can disconnect a subtree, leading to a topologically incorrect segmentation with a tiny decrease in the Dice loss [20]. Similarly, small over-segmentations of vessel trunks increase the Dice score without impacting the clinically important measure of network connectivity. This mismatch of volumetric and topology metrics has led to the invention of topologically sensitive loss functions.

2.4.1 Centreline Dice (clDice)

Shit et al. [20] proposed centreline Dice (clDice), a similarity index between the intersection of segmentation masks and their morphological skeleta, comprising two terms: topology precision (tprec) which is the proportion of predicted skeleton covered by the ground-truth mask, and topology sensitivity (tsens) which is the proportion of ground-truth skeleton covered by the predicted mask. The authors showed that for binary 2D/3D segmentations, clDice equal to 1 ensures topological equivalence up to homotopy. They also introduced a differentiable version of clDice for training, soft-clDice, which substitutes the hard morphological skeleton with an iteratively eroded probability map skeleton, and a standard segmentation loss, with a weighting factor α. Testing on five public datasets including vessels, roads, and neurons, both 2D and 3D, training soft-clDice (α range 0.1-0.5) yielded improved connectivity, graph similarity and Dice scores. Soft-clDice is efficient, with the loss computation for a 1024×1024 image at batch size 4 taking only 0.11 seconds per iteration longer than Dice.

The case for clDice should be interpreted with caution. The initial research [20] shows improvements on five diverse datasets but does not include hepatic vessels. Crucially, the study employs models lacking deep supervision, a component of nnU-Net, that, as we find (Section 3.5) may not play well with clDice's gradient signal at coarse decoder resolutions. Further, Kirchhoff et al. [23] and Shi et al. [24] point out specific issues relevant to hepatic vessel segmentation: clDice's uniform weighting of all skeleton voxels ignores the large diameter variation between portal trunk and peripheral branches; and clDice's GPU memory consumption scales poorly with 3D patch size. We nevertheless choose clDice as the most studied and theoretically sound choice at present, observing that a more principled alternative, cbDice [24], could be applied if its 3D implementation were available. We use clDice as an evaluation metric and as a loss function (coupled with Dice+CE), with additional detail on the training process in Section 3.5.

2.4.2 Persistent Homology Approaches

A different class of topology-related losses are based on persistent homology. Hu et al. [21] were the first to propose a differentiable loss enforcing the correct Betti numbers - the numbers of connected components and holes - which they showed to improve metrics that depend on topology on several datasets, but at a significantly higher computational cost than clDice. Clough et al. [22] developed a similar method that eliminates the need for ground-truth topology labels by including prior knowledge of the target structure's expected Betti numbers - appealing for semi-supervised learning, but dependent on consistent topology across patients, which is not the case for hepatic vessels, where the number of branches varies widely. Byrne et al. [25] generalized persistent homology losses to multi-class 3D segmentation, using cubical complexes and parallel processing, enabling their use in volumetric cardiac MRI, and offering a promising example that could be applied to future multi-class vessel segmentation of portal and hepatic veins.

2.4.3 Recent Alternatives to clDice

There are two recent approaches that target limitations of clDice relevant to this work. Kirchhoff et al. [23] introduced the Skeleton Recall Loss, which performs skeleton overlap computation on the CPU instead of the GPU, enabling avoidance of clDice's memory bottlenecks on large 3D volumes, and more than 90% reduction in computation cost. Importantly, the Skeleton Recall Loss also enables multi-class segmentation (unavailable in standard clDice), and so represents an exciting avenue for future research, including for separating portal veins from hepatic veins. Shi et al. [24] proposed cbDice, an extension of clDice that considers local vessel radius and boundary information. Standard clDice assigns equal weight to all skeleton voxels - a missed peripheral capillary and a missed major portal trunk both contribute equally to the loss - whereas cbDice takes vessel diameter into account, such that the loss is weighted by vessel size - an important consideration for hepatic vessel segmentation, where portal trunks and peripheral branches vary in diameter by over an order of magnitude.

2.5 Hepatic Vessel Segmentation: State of the Art

Having reviewed the general methodological building blocks, this section focuses on methods specifically targeting hepatic vessel segmentation and positions our approach within the current body of work.

2.5.1 Connectivity-Aware Approaches

Li et al. [1] plug a graph attention network (GAT) into 3D U-Net for hepatic vessel segmentation on MSD Task08, modeling vascular connectivity as a graph constructed from ground-truth vessel labels alone. The GAT plug-in added connectivity constraints as a multi-task learning branch only used during training-during inference time only the original U-Net is applied without increased runtime. Zhang et al. [2] took graph attention a step further with a graph-attention guided diffusion model that specifically addressed both connectivity and completeness at multiple scales and could effectively target smaller vessels. This model significantly outperformed eight other top-performing networks including nnU-Net and Swin UNETR on both 3D-IRCADb-01 and LiVS data. However, a multi-pass denoising process required by diffusion models dramatically increases runtime compared to single-pass networks such as nnU-Net.

To overcome scale imbalance Sadikine et al. [3] decomposed vessel tree to different scale levels using clustering and employed scale-specific auxiliary tasks and contrastive learning to distinguish vessel calibers. A complementary paper [4] proposed jointly multi-prior encoding by feeding shape and topological priors to a common encoder for liver vessel segmentation on 3D-IRCADb, showing that a fusion of these priors performed better than either one alone.

2.5.2 Direct Precedents

Our approach is closest in spirit to Nasser et al. [7]. They presented a set of experiments on liver vessels with both clDice and morphological skeletonisation losses and used connected-component and skeleton-based post-processing methods to generate multi-class outputs differentiating portal and hepatic veins. Their conclusions highlight the value of comparing different loss functions: they found that while differentiable skeletonisation improves skeleton quality, it did not improve segmentation performance, and that clDice training yielded variable results. Their results agree well with ours; as will be discussed in Chapter 4, while our clDice-trained model increased topology sensitivity over the Dice+CE baseline at the expense of a lower volumetric Dice, and our connected-component post-processing decreased vessel Dice by removing spurious but real vessel fragments. Garret et al. [5] experimented with concatenating vesselness filter outputs as inputs to a U-Net. On the IRCAD dataset, this approach provided an average +9% improvement in small vessel Dice over a baseline U-Net, but a degradation in large vessel performance as filter output saturated. Wu et al. [6] demonstrated a 3D Swin-Transformer network with voxel-wise rather than patch-wise embedding, IBIMHAV-Net, for liver vessels on the 3D-IRCADb dataset and reported average Dice of 74.8% and sensitivity 77.5%; however, evaluation was done on a statistically unreliable set of only 4 test images.

2.5.3 Published Benchmarks on MSD Task08

Table 1 gives a summary of published results for MSD Task08, along with our own calculated result. Note that the published methods were compared against the ground truth test set of 140 cases on the MSD challenge platform; our own number was derived from internal testing on a 20-case validation set, separate from training. This number is therefore not comparable directly and should be regarded as an internal validation, rather than competitive result. Nevertheless, it is interesting to see how it compares relatives to others.

A significant trend that one sees in published results is the trade-off between vessel and tumor performance; when high Dice is obtained for the vessels, Dice for the tumor becomes low and vice versa.

Table 1. *Segmentation performance on MSD Task08 (Hepatic Vessel). Note: highlighted rows (yellow) are our results on a 20-case internal validation split and are NOT directly comparable to published results, which use the official 140-case test set — a larger and more representative evaluation set evaluated through the MSD challenge platform. All relative comparisons in this work are therefore made against our own axial baseline only. ★ = best average Dice among our methods.*

Method	Year	Vessel Dice	Tumour Dice	Avg Dice	Eval Set
Kim et al.	2021	62.34%	68.63%	65.49%	Official test (140 cases)
Trans VW	2021	65.80%	71.44%	68.62%	Official test (140 cases)
nnU-Net (official)	2021	81.64%	52.78%	67.21%	Official test (140 cases)
DiNTS	2022	81.02%	55.35%	68.19%	Official test (140 cases)
Swin UNETR	2022	81.85%	58.21%	70.03%	Official test (140 cases)
Universal Model (CLIP)	2023	82.84%	62.33%	72.59%	Official test (140 cases)
Ours: MV Fusion (Dice+CE)	**2026**	**65.98%**	**63.89%**	**64.93%**	**Val split (20 cases) – NOT directly comparable to above**
Ours: MV Fusion (clDice)	**2026**	**62.84%**	**66.34%**	**64.59%**	**Val split (20 cases) – NOT directly comparable to above**

2.6 Ensemble and Fusion Methods in Medical Segmentation

The EMMA model proposed by Kamnitsas et al. [37] for brain tumour segmentation won the BraTS 2017 competition by averaging the outputs of a set of heterogeneous CNN architectures. The reasoning is that heterogeneous models tend to be more robust than any one model alone because independent errors from each CNN structure cancel each other out – although each of the individual networks employed was of relatively simple design and was not trained specifically for the task, their average predictions generalized well to the novel test data set. We directly utilize this reasoning for our majority voting ensemble; in combining the predictions from models trained using each orientation of data, we can expect to see the removal of orientation-dependent errors when these predictions disagree.

Weiler et al. [39] employed a similar method; they used a rater-aware training approach in nnU-Net, called MLV2-Net, to segment meningeal lymphatic vessels. They employ strategies such as majority voting with weights to average the models which achieved a Dice of 0.806 to the human gold standard. Their results support the viability of training an nnU-Net and majority voting for vessel segmentation specifically.

Learned strategies of merging CNN outputs provide a more adaptive alternative to predefined rules. Dang et al. [38] used a global learning particle swarm optimisation to learn fusion weights, and in another publication [45], they proposed a two-stage ensemble which trains a second-layer network using the output of first-layer models as augmented features, achieving performance beyond simple averaging on cardiac segmentation benchmarks. In this work, however, our learned CNN-based fusion achieved worse performance (61.73%) than majority voting (64.93% average Dice), showing similar sensitivity to overfitting as previously found; a greater volume of training data is needed for learned methods than simple voting approaches.

Post-hoc refinement techniques allow further improvement to segmentation. Carneiro-Esteves et al. [42] used a learned post-processor to reconnect broken vessel fragments and demonstrated an increase in connectivity for both 2D and 3D segmentations. Ye et al. [44] use graph-based curvilinear clustering within VSR-Net for the merging of subsection breaks. Both approaches allow for the restoration of connectivity to the segmented structures, though it is worth noting that post-hoc refinement with connected-component filtering alone risked the removal of genuine fragments, as demonstrated by Nasser et al. [7].

2.7 Summary and Research Gap

We build on three observations from this survey. We observe that hepatic vessels are challenging to segment because of class imbalance, their filigree structure and inter-patient variability. Although multiplanar networks have been shown to improve performance on a range of segmentation targets by capturing complementary spatial context across orthogonal planes, this has rarely been explored for vessels. On the other hand, topological losses like clDice provide theoretically motivated approaches to enhance the connectivity of vessel networks; but their combination with nnU-Net's deep supervision, and their effectiveness when combined with multiplanar representations, has been little explored.

These three themes combine to a single structural point: The two modes of failure (orientation-specific blind spots and topological breakage) of single-orientation Dice+CE training of hepatic vessels can be separately addressed by multiplanar fusion and clDice, so they are natural partners. We test this hypothesis here: we first train orientation-specific nnU-Net v2 models on the MSD Task08 hepatic vessel dataset, reoriented in the axial, sagittal and coronal planes, and then fuse them via majority voting, and compare their performance with clDice training at the level of each orientation. This is the first systematic investigation of multiplanar nnU-Net fusion with topological training on this task, to our knowledge.

Chapter 3. Methodology

This chapter describes the complete experimental pipeline, from dataset preparation through evaluation. All experiments use the nnU-Net v2.6.2 framework with the 3D_fullres configuration. An overview of the pipeline is illustrated in Figure 2.

Figure 1 *contrasts the conventional single-orientation pipeline with the proposed multiplanar approach. The key difference is that instead of training one axial model, three orientation-specific models are trained and their predictions combined — allowing each model to resolve vessel branches that may be ambiguous or poorly resolved in the other views.*

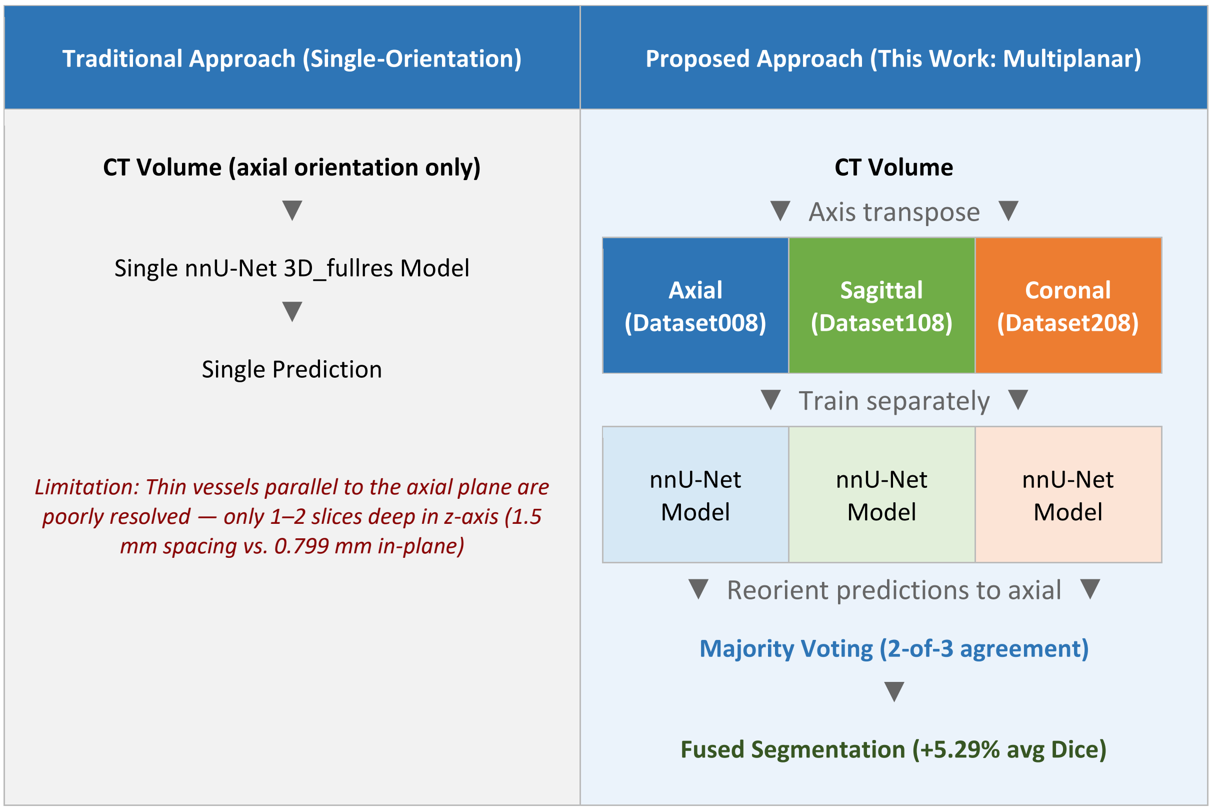

Figure 1. *Graphical abstract. Left: the conventional single-orientation pipeline processes CT volumes from the axial view only, missing vessel branches that are poorly resolved in that plane due to anisotropic CT spacing (1.5 mm axial vs. 0.799 mm in-plane). Right (this work): three independent nnU-Net 3D_fullres models are trained on axial, sagittal, and coronal reorientations of the same data. At inference, predictions are transformed back to axial space and combined via majority voting — a voxel is labelled as vessel or tumour if at least 2 of 3 models agree — improving average Dice by +5.29% (p = 0.0045).*

Experiment	Research question	Hypothesis	Section	Finding
Exp 1 MV Fusion (Dice+CE) vs Baseline	RQ1: Does multiplanar fusion improve accuracy over a single axial model?	H1: Fusion compensates for orientation-specific errors	4.1–4.2	✓ Confirmed +5.29% avg Dice, p = 0.0045
Exp 2 Fusion strategy comparison	RQ3: Which fusion strategy performs best?	H3: Simple voting outperforms learned fusion at this data scale	4.2	✓ Confirmed MV voting 64.93% > CNN fusion 61.73%
Exp 3 clDice loss (single axial model)	RQ2: Does clDice training improve vessel topology?	H2: clDice improves topology at potential cost to volumetric Dice	4.3	~ Partial Higher sensitivity, lower vessel Dice. Topology mixed.
Exp 4 MV Fusion (clDice)	RQ1 + RQ2: Do fusion and clDice provide complementary benefits?	H1 + H2 are complementary when combined	4.4	✓ Confirmed Best topology (clDice 70.47%) + best tumour Dice (66.34%)
Exp 5 Connected-component post-processing	—	Expected to remove spurious small predictions	4.5	✗ Negative result Hurts vessel Dice (−1.05%, p = 0.0107)

Table 2. *Mapping of each ablation experiment to its corresponding research question, hypothesis, results section, and outcome. Green = confirmed hypothesis, yellow = partial, orange = negative result.*

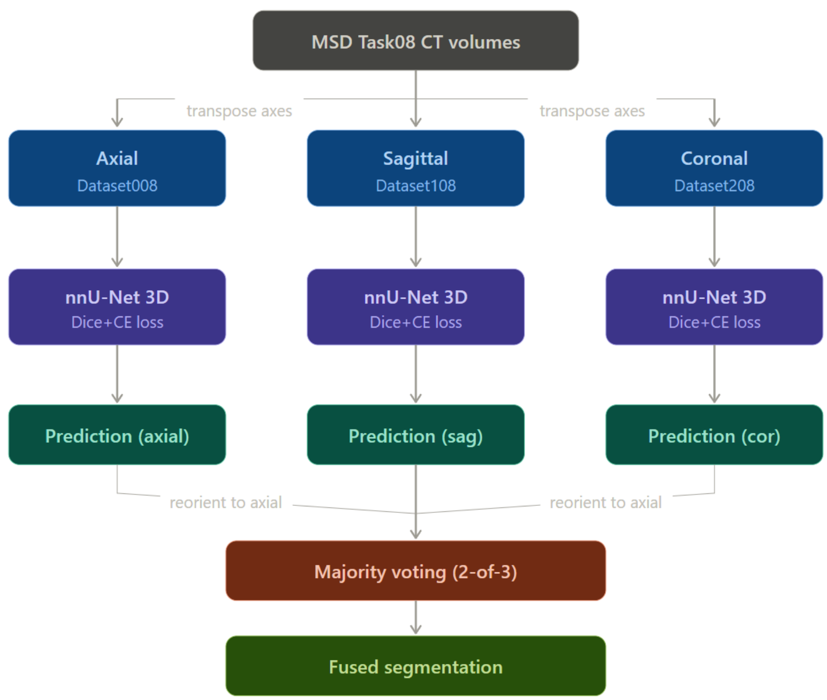

Figure 2. (1) Original axial CT volume → axis transposition → three datasets (Axial/Sagittal/Coronal); (2) Three independent nnU-Net 3D_fullres models trained separately; (3) Inference → reorient predictions back to axial space → majority voting fusion → final segmentation.

3.1 Dataset

For our experiments we use the Medical Segmentation Decathlon (MSD) Task08 - Hepatic Vessel dataset [40, 41]. This dataset includes 443 portal-venous phase CT scans annotated at the voxel-level for 2 foreground labels, namely hepatic vessels (label 1) and hepatic tumours (label 2). The dataset was converted into nnU-Net format (as Dataset008HepaticVessel, while maintaining the original labels). The labeled MSD training set contains 303 annotated cases. The 140 cases left in the dataset constitute the official MSD challenge test set, for which ground truth labels are unavailable (prediction on these require submission to the MSD challenge server). As we aimed for rapid evaluation on numerous ablation studies through local experimentation, we instead used 20 of the 303 labeled cases as a held-out test set. These 20 held-out cases (IDs: hepaticvessel_429 to hepaticvessel_458) were excluded from training and cross-validation. We selected the last 20 cases alphabetically and excluded them from the dataset prior to nnU-Net's default cross-validation split (leaving 283 cases for training). This split into 242/41 training/validation images was generated by nnU-Net for its fold 0 on the 283 training images. Therefore, none of the 20 held-out test images were used for checkpointing or any decision about the model training and there is no data leakage into the training process.

The volumes have an approximate median image size of (150, 512, 512) voxels following the resampling to target spacing of (1.5, 0.799, 0.799) mm. The z-axis (axial) spacing is roughly twice the in-plane (x, y) spacing. This is a typical characteristic of clinical abdominal CT scans and a motivation for our multiplanar approach; on the axial slices with coarser spacing the vessels traveling parallel to the slice might span just 1-2 slices, rendering detection difficult without considering multiplanar information.

3.2 nnU-Net Configuration and Baseline Training

The baseline model is an off-the-shelf nnU-Net v2 3D_fullres PlainConvUNet, trained on the axial data (Dataset008). The architecture is composed of 6 stages in the encoder-decoder network with number of feature maps of (32, 64, 128, 256, 320, 320), 3D convolutions with a kernel size of 333 across all stages, instance normalisation and LeakyReLU activation. Strides are (1,1,1), (2,2,2), (2,2,2), (2,2,2), (2,2,2), (1,2,2) thus covering a receptive field of the entire patch. Training takes place with the default Dice + cross-entropy (Dice+CE) loss with deep supervision, SGD optimiser with an initial learning rate of 0.01 and polynomial decay. The model uses nnU-Net's default augmentation pipeline of rotation, scaling, elastic deformation, gamma correction and mirroring. The model is trained for around 250 epochs on fold 0 and the checkpoint with best validation Dice is saved.

3.3 Multiplanar Data Preparation

To generate sagittal and coronal training sets we perform axis transposition on each volume and its corresponding label of the original axial dataset:

Sagittal (Dataset108): Transposes axes from (X,Y,Z) to (Z,Y,X). This is equivalent to rotating the volume such that the new "axial" axis corresponds to what the sagittal view was.

Coronal (Dataset208): Transposes axes from (X,Y,Z) to (Y,Z,X), such that the coronal view corresponds to the new "axial" axis.

The affine matrices for NIfTI files are accordingly modified to preserve correct spatial location of points. Note that no additional processing is done on images and labels. They are essentially same as in original volume, only their axis ordering is different. Each orientation's model thus learns to perceive the same anatomy from a different angle.

All the three reoriented datasets were processed and trained with nnU-Net individually with identical preprocessing (fingerprinting, spacing examination, and patch selection) resulting into a separate 3D_fullres model for each. These three models share identical architecture configuration, 6 stage PlainConvUNet with the same patch size (64,192,192) due to the fact that the reoriented volumes had very similarmedianshape and spacings.

3.4 Fusion Strategies

In inference, all 3 models predict images with their own orientation. The predictions from the sagittal and coronal models are resampled back into the axial coordinate system via reversal of the axis permutations during dataset construction. All 3 prediction volumes are now in the same coordinate system and can be combined on a voxel-by-voxel basis.

We experimented with four fusion schemes:

Majority Voting: at each voxel, the prediction from at least 2 out of 3 models is chosen. For 3 models and 3 classes (background, vessel, tumour), this is equivalent to choosing the mode.

Average Voting: The prediction maps from the 3 models were averaged and rounded to obtain the final prediction. For binary-like predictions from 3 models, this yields the same results as majority voting - we confirmed this empirically in experiments.

Weighted Voting: The three models' predictions were averaged but weighted according to their validation Dice scores prior to averaging. Using approximate weightings of 0.73/0.72/0.72 (axial/sagittal/coronal) produced identical results to majority voting. The weights were roughly equal as a consequence of the three models performing very similarly on the validation set.

Learned Fusion (CNN): A simple 3D CNN takes the stack of the three orientation model's one-hot encoded predictions as input (9 input channels; 3 orientations * 3 classes) and learns to produce the fused image. The network consists of two 3x3x3 convolutional layers; the first has 16 output channels with batch normalisation and ReLU activation, and the second has 3 output channels. This yields approx. 5,000 parameters. We trained using cross-entropy on the Adam optimiser (lr=1e-5) for 10 epochs, with a batch size of 2, training on randomly sampled 64x64x64 image patches drawn from the 283 non-test training cases. A low capacity learned fusion was desired, in order to avoid overfitting given the relatively small training data size; this choice thus permits examination of whether any significant improvement can be gained from learning-based fusion within the given constraints.

3.5 clDice Loss Integration

To train with clDice as a loss, we modified the nnU-Net trainer class (nnUNetTrainerClDice) to inherit from the default nnUNetTrainer. The trainer uses a warm-up-ramp training schedule:

- **Epochs 0-49**: Dice+CE loss (same as baseline). This allows the model to converge to a good segmentation before the topology regularisation begins.

- **Epochs 50-149**: clDice weight linearly increased from 0.0 to 0.1. This avoids instability from loss surface changes.

- **Epochs 150-249**: clDice weight set to 0.1. The total loss is: L = L_Dice+CE + 0.1 × L_clDice.

This low weight was selected after preliminary experiments with higher clDice weights showed unstable training. A weight of 0.5 (as used in the original clDice paper [20]) led to diverging loss; 0.2 removed the divergence but still led to unstable validation curves. We believe this is due to the interaction between the clDice gradient computation and nnU-Net's deep supervision, which involves multiple losses from the intermediate scales of the decoder (not topology-aware) and the full-resolution clDice (topology-aware). Lowering the weight to 0.1 and introducing the clDice at epoch 50 eliminated the instability while still offering a topology-aware loss. The cldice training schedule is shown in **Figure 3.**

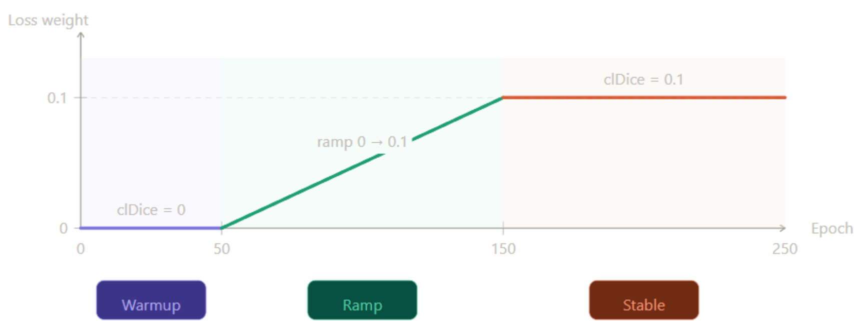

Figure 3. *clDice loss weight schedule used during training. Epochs 0–49: warmup phase, standard Dice+CE loss only (clDice weight = 0). Epochs 50–149: ramp phase, clDice weight linearly increased from 0.0 to 0.1. Epochs 150–249: stable phase, clDice weight fixed at 0.1. This gradual introduction was necessary to prevent interference with nnU-Net's deep supervision mechanism at higher weights.*

The nnUNetTrainerClDice class overrides the __init__ method using inspect.signature to handle nnU-Net's dynamic constructor and computes soft-clDice using the skimage.morphology.skeletonize_3d function for the morphological skeleton. (see **Figure A5**, Appendix A for the training curve confirming stable convergence)

3.6 Post-Processing

The final step post the fused predictions consists of filtering connected components. Components which are smaller than 50 voxels for vessels or 100 voxels for tumours are discarded and small interior holes smaller than 500 voxels in any remaining structure are filled. In the rare cases of overlapping structures, the tumour label wins.

The value of the 50 voxel vessel threshold was chosen without the availability of a validation set for threshold optimization, and should be viewed as a design limitation rather than an optimal parameter. At the target spacing of $1.5 \times 0.799 \times 0.799$ mm, each voxel occupies approximately 0.958 mm³, so a 50-voxel component corresponds to ~47.9 mm³. By contrast, a terminal peripheral branch of 1 mm diameter and 5 mm length has a cylindrical volume of $\pi \times 0.5^2 \times 5 \approx 3.9$ mm³, equivalent to roughly 4 voxels at this spacing; even accounting for partial-volume labelling, such a branch is unlikely to exceed 10–15 voxels [1, 6]. The 50-voxel threshold therefore substantially exceeds the volume of genuine peripheral branches, meaning that real vessel fragments are systematically removed alongside noise. Consequently, the negative result at post-processing reported in Section 4.5 is an expected consequence of this over-aggressive threshold rather than a fundamental limitation of connected-component filtering as a technique. A 100-voxel tumour threshold corresponds to ~95.7mm³, which for a spherical lesion implies a diameter of ~5.7 mm. Small (<1cm) hepatic lesions are typically considered ambiguous in the CT staging of cancer [26], and the threshold seems reasonable. Future approaches could employ more principal threshold selection strategies; e.g., lower bound dictated by smallest component observed in ground-truth training labels.

3.7 Evaluation Metrics

All methods were tested on the same 20 unseen test cases using the following measures:

Dice Similarity Coefficient (DSC) - the standard volumetric overlap metric, measured per-class (vessel and tumour) and then averaged.

Centreline Dice (clDice) - a topology-preserving metric calculated on the intersection of predicted and ground-truth skeletons. This is reported along with its components - topology precision (tprec) and topology sensitivity (tsens).

95th Percentile Hausdorff Distance (HD95) - a surface distance metric to assess the accuracy of the boundary, reported in millimeters.

Sensitivity (Recall) - the fraction of positive voxels from the ground-truth which are correctly predicted.

Specificity - the fraction of negative voxels from the ground-truth which are correctly predicted, for all methods it is greater than 99.9% due to the extreme class imbalance.

Precision - the fraction of predicted positive voxels which are actually positive.

3.8 Statistical Analysis

To determine the statistical significance of the differences between methods, a Wilcoxon signed-rank test (two-sided) is used which is appropriate for n=20, a small sample size, with no assumption of normal distribution of the scores, and uses a pair-wise test: *** indicates $p < .001$, ** indicates $p < .01$, * indicates $p < .05$, and ns is used if $p \geq 0.05$. Effect sizes (rank-biserial correlation r) are presented with the p-values, in order to make interpretation possible regardless of the size of the sample.

There are two limitations of the statistics used which are needed to be mentioned. Firstly, with the test set of n=20, there is limitation on the power of the tests, so differences of less than about 3-4 percent points should be considered not significant, since the effect could still be there, but we are only seeing a difference that is potentially underpowered and that a Type II error may have been made, and secondly the use of eight pairwise comparisons in Table 12 makes it a question of the effect of the multiple comparisons.

3.9 Hardware and Software

All model training and inference is performed on Google Colab Pro using NVIDIA T4 (16GB), A100 (80GB), or H100 (80GB) GPUs depending on availability. Training time is approximately 12–14 hours per model. Colab session disconnections are handled via nnU-Net's checkpoint resumption (--c flag), with all data stored on Google Drive for persistence. The anti-disconnect JavaScript script is used to maintain sessions during long training runs.

Software versions: nnU-Net v2.6.2, PyTorch (Colab default), Python 3.12, nibabel for NIfTI I/O, scikit-image for skeletonisation, scipy for statistical testing.

3.10 Reproducibility

All experiments are implemented in a single Google Colab notebook that contains the complete pipeline from dataset preparation through final evaluation. Trained model checkpoints, per-case prediction NIfTI files, and all evaluation CSVs are stored on Google Drive and are available upon request. The custom nnUNetTrainerClDice trainer class is self-contained and can be installed into any nnU-Net v2 installation by placing the Python file in the nnunetv2/training/nnUNetTrainer/ directory. All results reported in Chapter 4 were verified by recomputing Dice scores directly from the NIfTI files on disk, with cross-checks between multiple CSV sources to detect stale or inconsistent values.

Chapter 4. Results

This chapter presents the experimental results across all ablation experiments. All values are computed from the actual NIfTI prediction files on disk and verified for consistency.

4.1 Baseline and Single-Orientation Models

Table 3 shows the Dice scores for each single-orientation model trained with standard Dice+CE loss.

Table 3. Dice scores for single-orientation models (Dice+CE loss, 20 test cases).

Model	Vessel Dice	Tumour Dice	Average Dice
Axial (Baseline)	66.03%	53.26%	59.65%
Sagittal	62.83%	57.29%	60.06%
Coronal	64.00%	51.54%	57.77%

Single orientation: Axial achieves the highest vessel Dice (66.03%), Sagittal achieved highest tumour Dice (57.29%) and Coronal had the worst results on both classes. Neither view is best for both classes of anatomies.

4.2 Fusion Strategy Comparison

Table 4 compares all fusion strategies applied to the three Dice+CE models.

Table 4. Fusion strategy comparison (Dice+CE models, 20 test cases).

Method	Vessel Dice	Tumour Dice	Avg Dice	Δ Baseline
Majority Voting	65.98%	63.89%	64.93%	+5.29%
Average Voting	65.98%	63.89%	64.93%	+5.29%
Weighted Voting	65.98%	63.89%	64.93%	+5.29%
Learned Fusion (CNN)	62.81%	60.66%	61.74%	+2.09%

Overall Majority voting gives the best performance, 64.93% average Dice, with the most dramatic increase on tumour Dice (+10.63% over baseline). Interestingly the 3 voting strategies all produce exactly the same results, Majority voting, average voting and weighted voting. This is because when having three binary voters with approximately equal performance (validation Dice approximately 0.73/0.72/0.72) then weighted averaging is exactly equal to average voting and when voting the final result, the outcome is exactly the same discrete label decision as majority voting. The CNN learned fusion gives a worse result than the majority voting at 61.73% compared to 64.93%. This is probably because the very minimal architecture which has been used - less than 5000 parameters and only trained on 64 patches - is not sufficient to learn any intra-orientation pattern information other than that obtained already by the basic majority voting approach.

(see Figure A1, Appendix A for a qualitative comparison on the easy case hepaticvessel_429; Figure A2, Appendix A for a medium-difficulty case hepaticvessel_437; Figure A4, Appendix A for a second medium-difficulty case hepaticvessel_443, demonstrating improved tumour boundary delineation across all fusion methods)

4.3 clDice Loss Training

Table 5 shows results for the axial model trained with clDice loss compared to the standard baseline.

Table 5. Axial clDice model vs baseline (20 test cases).

Model	Vessel Dice	Tumour Dice	Avg Dice	Vessel clDice
Axial Baseline (Dice+CE)	66.03%	53.26%	59.65%	70.05%
Axial clDice	60.40%	61.15%	60.77%	69.45%

The trade-off observed for the clDice-trained model is a decrease of 5.63% in the Dice value of vessels (significant with a value of $p = 0.0094$) and an increase of 7.89% in the Dice value of tumors (not significant with a value of $p = 0.1365$). There is no change (almost) in the Dice value for the vessels in terms of their clDice value (topology) 69.45% and 70.05% respectively. The reason why this happens becomes evident by inspecting the confusion metrics; for vessels, the sensitivity has improved (78.44% compared to 71.61%) and the precision has degraded (52.74% compared to 66.45%) with a higher sensitivity meaning that more vessel voxels are being detected (though with an increased number of false positives), hence the model is more likely to produce false positives. The model now behaves more like a recall-driven model. This recall-driven characteristic of the model can be attributed to the focus on center line coverage that clDice implements. The loss will thus penalize those missing voxels on the centreline and hence the model tends to over-predict in this case.

4.4 Multiplanar clDice Fusion

All three orientation models were retrained with the clDice loss schedule and fused via majority voting. Table 6 presents the results.

Table 6. Experiment 3 — clDice models with majority voting fusion (20 test cases).

Model	Vessel Dice	Tumour Dice	Avg Dice
Axial clDice	60.40%	61.15%	60.77%
Sagittal clDice	62.81%	53.21%	58.01%
Coronal clDice	60.80%	56.06%	58.43%
MV Fusion (clDice)	**62.84%**	**66.34%**	**64.59%**

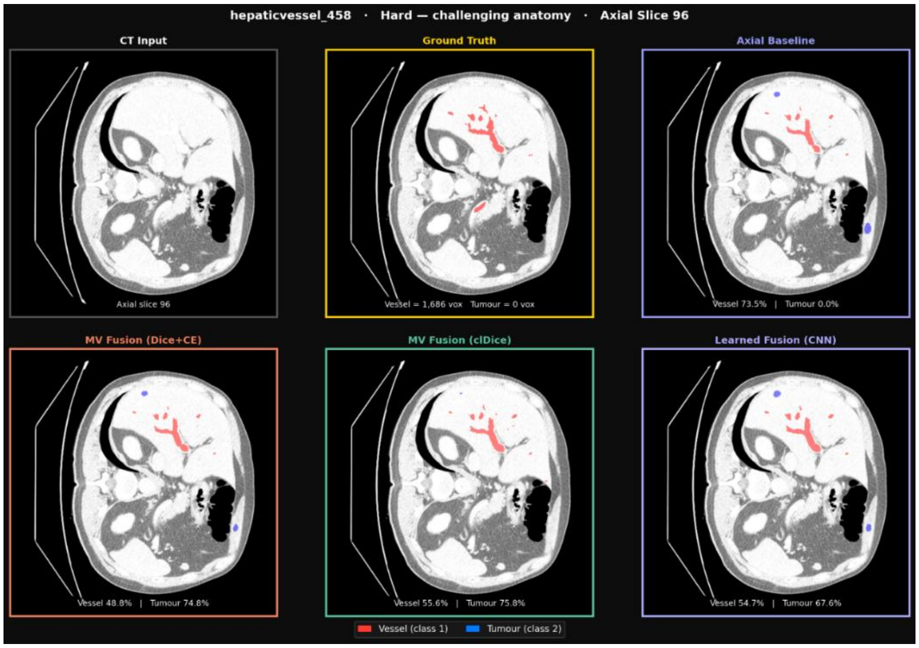

Figure 4. *Qualitative comparison on hepaticvessel_458 (hard case, challenging anatomy), axial slice 96. MV Fusion (clDice) recovers tumour regions missed by the axial baseline, consistent with its improved tumour Dice (+18.21% in this case).*

Figure 4 provides a qualitative result in hard case (hepaticvessel_458) and illustrates that MV Fusion (clDice) recover tumor regions failed by axial baseline. In terms of performance, the clDice fusion gains an average Dice of 64.59%, +4.95% compared to axial baseline, and the tumor Dice value is highest of all experiments (66.34%). Improvement against the axial baseline is significant (p=0.0400). The gap between MV Fusion (clDice) and MV Fusion (Dice+CE) is not significant (p=0.6742). The two best methods are statistically on par in terms of overall performance.

Notably, the clDice fusion achieves the best topology score of any method:

Table 7. Topology metrics (vessel class, 20 test cases).

Method	Vessel clDice	Topology Precision	Topology Sensitivity
MV Fusion (Dice+CE)	69.41%	69.35%	68.64%
MV Fusion (clDice)	70.47%	70.11%	75.65%

The clDice fusion shows higher topology sensitivity (75.65% vs 68.64%), confirming that the clDice-trained models are better at capturing vessel centrelines. However, this comes at a slight cost to topology precision.

4.5 Post-Processing Ablation

Connected-component filtering was applied to both best fusion methods. Table 8 shows the results.

Table 8. Post-processing results (20 test cases).

Method	Vessel Dice	Tumour Dice	Avg Dice
MV Fusion (Dice+CE)	65.98%	63.89%	64.93%
MV Fusion (Dice+CE) + PP	64.93%	64.01%	64.47%
MV Fusion (clDice)	62.84%	66.34%	64.59%
MV Fusion (clDice) + PP	62.04%	66.60%	64.32%

Post-processing consistently hurts vessel Dice (−1.05% for Dice+CE, −0.80% for clDice) while providing marginal tumour improvement. For the Dice+CE fusion, the vessel drop is statistically significant ($p = 0.0107$). This is a negative result: connected-component filtering removes small vessel fragments that, in many cases, are genuine peripheral branches rather than false positives. This finding aligns with observations by Nasser et al. [7] on the difficulty of post-processing vessel segmentations without damaging real anatomy.

4.6 Topology Evaluation (clDice Metric)

Table 9 summarises the clDice topology scores across all methods for the vessel class.

Table 9. Vessel topology metrics (20 test cases).

Method	Dice	clDice	Gap	Interpretation
Axial Baseline	66.03%	70.05%	−4.02%	Good topology
Sagittal	62.83%	68.37%	−5.54%	Good topology
Coronal	64.00%	69.59%	−5.59%	Good topology
MV Fusion (Dice+CE)	65.98%	69.41%	−3.43%	Good topology
Axial clDice	60.40%	69.45%	−9.05%	More fragmented
MV Fusion (clDice)	62.84%	70.47%	−7.63%	Moderate

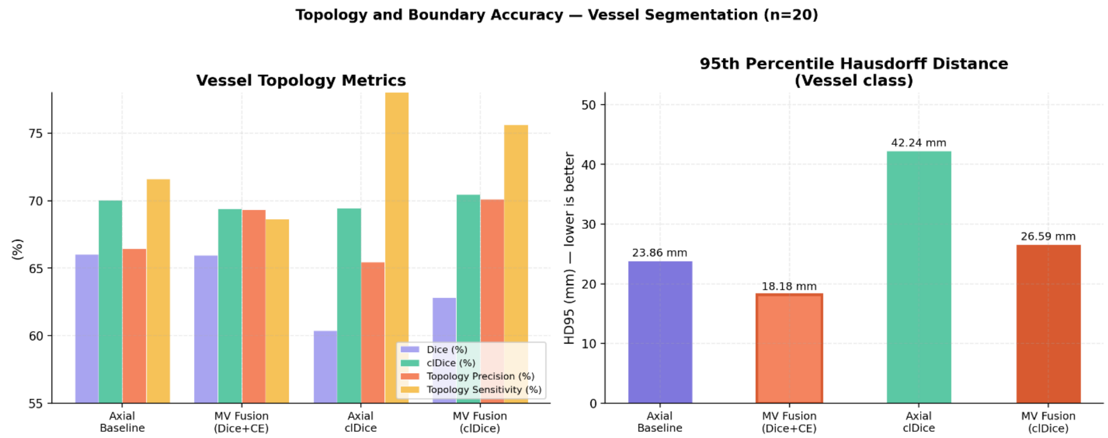

Figure 5. *Vessel topology metrics for key methods. Left: Dice, clDice, topology precision (tprec), and topology sensitivity (tsens). Right: 95th percentile Hausdorff distance (HD95, mm) — lower is better. Orange border = best result per metric.*

Figure 5 summarises vessel topology and boundary metrics across key methods. A multi-level axial comparison across three representative slices of hepaticvessel_429 is provided in **Figure A3** (Appendix A), demonstrating consistent segmentation quality across the full axial range. A negative gap (clDice > Dice) is expected for vessel segmentations and indicates that the model captures centreline structure better than volumetric extent. Larger gaps suggest more fragmented predictions where many small vessel fragments are detected but not at their full diameter. The clDice-trained models show larger gaps, consistent with their higher sensitivity but lower precision.

4.7 HD95 and Boundary Accuracy

Table 10 shows the 95th percentile Hausdorff distance for the fusion methods.

Table 10. HD95 in millimetres (20 test cases, lower is better).

Method	Vessel HD95	Tumour HD95
Axial Baseline	23.86	115.89
MV Fusion (Dice+CE)	18.18	43.74
MV Fusion (clDice)	26.59	58.81

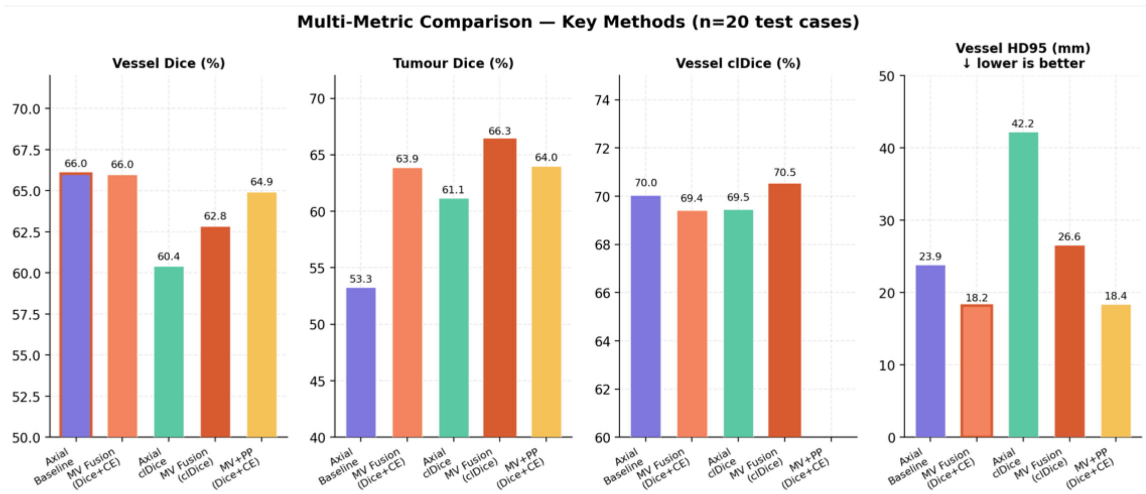

Figure 6. *Multi-metric comparison of key methods: Vessel Dice (%), Tumour Dice (%), Vessel clDice (%), and Vessel HD95 (mm). Orange border marks the best result per metric. N/A indicates metric not computed for that method.*

Figure 6 provides a multi-metric comparison across vessel Dice, tumour Dice, vessel clDice, and HD95. Majority voting with Dice+CE models achieves the best vessel HD95 (18.18 mm), indicating more accurate vessel boundaries. The clDice fusion has higher HD95 for vessels (26.59 mm), consistent with its more aggressive but less precise predictions. Both fusion methods substantially improve tumour HD95 over the baseline, with Dice+CE fusion achieving 43.74 mm versus 115.89 mm for the axial baseline.

4.8 Sensitivity, Specificity, and Precision

Table 11 summarises the confusion-matrix-derived metrics for key methods.

Table 11. Sensitivity, specificity, and precision (20 test cases).

Method	Class	Sensitivity	Specificity	Precision
Axial Baseline	Vessel	71.61%	99.97%	66.45%
Axial Baseline	Tumour	75.75%	99.96%	44.88%
MV Fusion (Dice+CE)	Vessel	68.64%	99.98%	69.35%
MV Fusion (Dice+CE)	Tumour	71.73%	99.98%	60.04%
MV Fusion (clDice)	Vessel	76.29%	99.95%	57.71%
MV Fusion (clDice)	Tumour	78.39%	99.97%	60.64%
Axial clDice	Vessel	78.44%	99.94%	52.74%

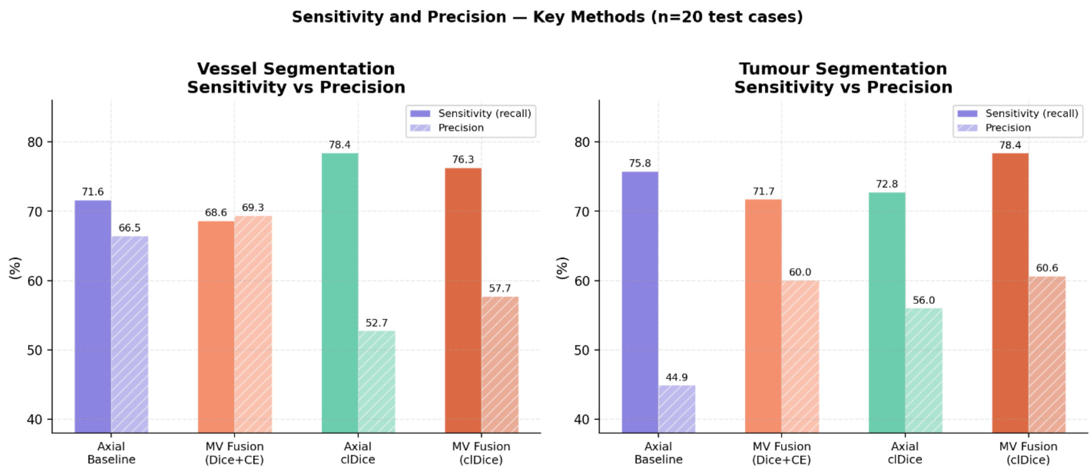

Figure 7. *Sensitivity (solid) and Precision (hatched) for vessel and tumour segmentation across key methods. MV Fusion (Dice+CE) improves precision at a small cost to sensitivity — the consensus filtering effect. clDice models show the inverse pattern, prioritising recall over precision.*

Figure 7 demonstrates the trade-off between sensitivity and precision for both vessels and tumours on the majority of methods. The majority vote provides a way to act as a consensus filter, as shown to be moderately beneficial in increasing precision for both vessels (69.35% compared to 66.45%) while only slightly decreasing vessel sensitivity (68.64% versus 71.61%). For tumors, the jump in precision is more impressive (60.04% versus 44.88%) to which the large gain in tumor Dice is attributed. For clDice, both vessel and tumor results flip the pattern and are found to perform more favorably in sensitivity, while suffering in precision.

4.9 Statistical Significance

Table 12 summarises the Wilcoxon signed-rank test results for key comparisons.

Table 12. Statistical significance (Wilcoxon signed-rank test, two-sided, n=20).

Comparison	Metric	Diff	p-value	Sig
MV Fusion (Dice+CE) vs Baseline	Vessel	−0.05%	0.8124	ns
MV Fusion (Dice+CE) vs Baseline	Tumour	+10.63%	0.0029	**
MV Fusion (Dice+CE) vs Baseline	Avg	+5.29%	0.0045	**
clDice Loss vs Baseline	Vessel	−5.63%	0.0094	**
clDice Loss vs Baseline	Tumour	+7.89%	0.1365	ns
clDice Loss vs Baseline	Avg	+1.13%	0.9854	ns
MV Fusion (clDice) vs Baseline	Avg	+4.95%	0.0400	*
MV Fusion (clDice) vs MV Fusion (Dice+CE)	Avg	−0.34%	0.6742	ns

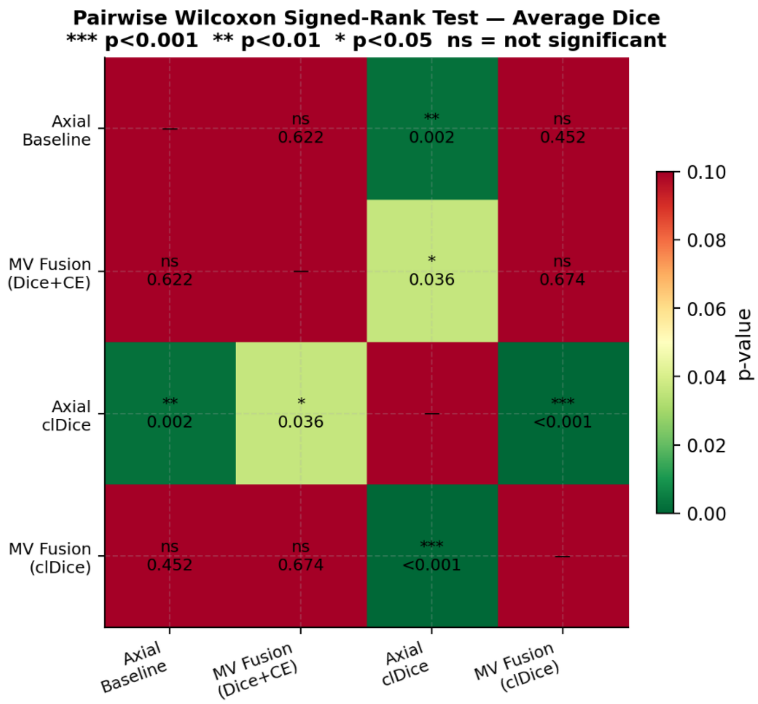

Figure 8. *Pairwise Wilcoxon signed-rank test p-value matrix (two-sided, n=20) for average Dice across key methods. Green = statistically significant difference (p<0.05); red = not significant. *** p<0.001, ** p<0.01, * p<0.05, ns = p≥0.05.*

The results for the pairwise significance can be viewed in **Figure 8**. The main and only important observation is that the increase due to majority voting (+5.29%) is significant (p=0.0045) which suggests that the increase is not simply due to random variation and the multiplanar approach truly yields a real increase. The biggest and best individual increase comes from the tumour (+10.63%, p=0.0029) and the best two fusion methods (MV (Dice+CE) and MV (clDice)) are statistically equal (p=0.6742).

4.10 Consolidated Ablation Table

Table 13 presents the complete ablation study.

Table 13. Consolidated ablation study (all methods, 20 test cases, computed from files on disk).

Method	Vessel	Tumour	Avg	Δ Baseline
Single Models				
Axial Baseline (Dice+CE)	66.03%	53.26%	59.65%	—
Sagittal (Dice+CE)	62.83%	57.29%	60.06%	+0.41%
Coronal (Dice+CE)	64.00%	51.54%	57.77%	−1.88%
Axial clDice	60.40%	61.15%	60.77%	+1.12%
Sagittal clDice	62.81%	53.21%	58.01%	−1.63%
Coronal clDice	60.80%	56.06%	58.43%	−1.22%
Fusion				
MV Fusion (Dice+CE) ★	65.98%	63.89%	**64.93%**	+5.29%
MV Fusion (clDice)	62.84%	**66.34%**	64.59%	+4.95%
Learned Fusion (CNN)	62.81%	60.66%	61.73%	+2.09%
Post-Processed				
MV Fusion (Dice+CE) + PP	64.93%	64.01%	64.47%	+4.82%
MV Fusion (clDice) + PP	62.04%	66.60%	64.32%	+4.67%

★ Best overall average Dice. The best tumour Dice is achieved by MV Fusion (clDice) at 66.34%. Average voting and weighted voting are omitted as they produce identical results to majority voting (verified from disk).

Figure 9 visualises the vessel, tumour, and average Dice scores across all ablation methods. **Figure 10** shows the per-case average Dice distribution as violin plots, confirming that case difficulty drives variance more than method choice**. Figure 11** traces the incremental ablation progression from axial baseline to best fusion method, with per-stage Δ over baseline**.**

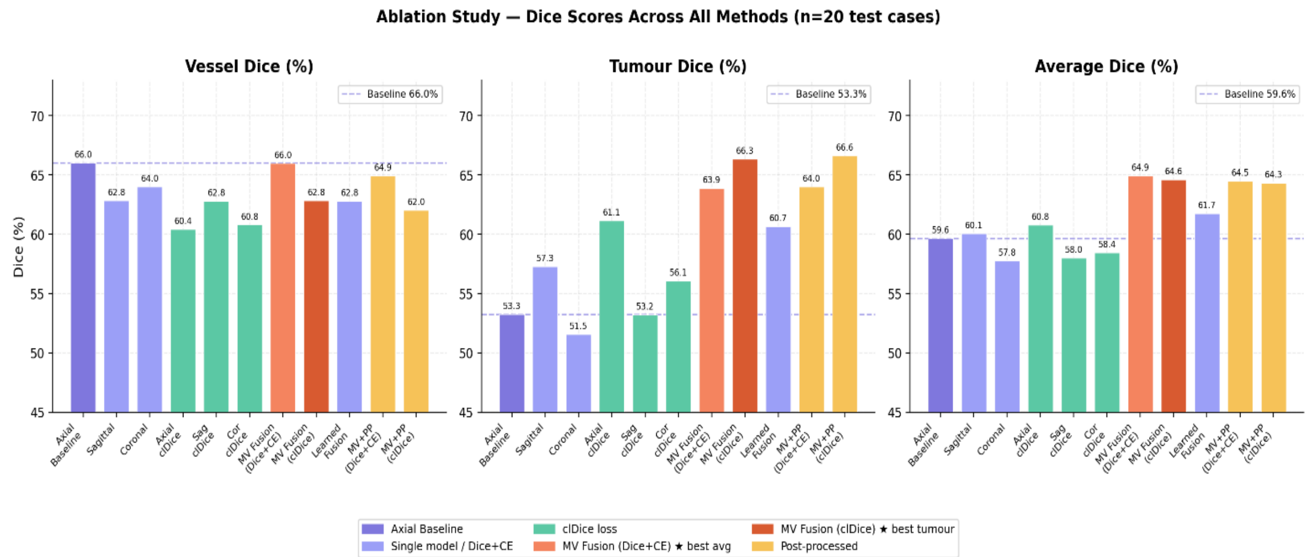

Figure 9. *Vessel, Tumour, and Average Dice scores across all ablation methods (n=20 test cases). Coloured bars distinguish single-orientation models (purple), fusion strategies (orange/teal), and post-processed variants (amber). Dashed line marks the axial baseline (59.65%). ★ = best average Dice (MV Fusion Dice+CE, 64.93%).*

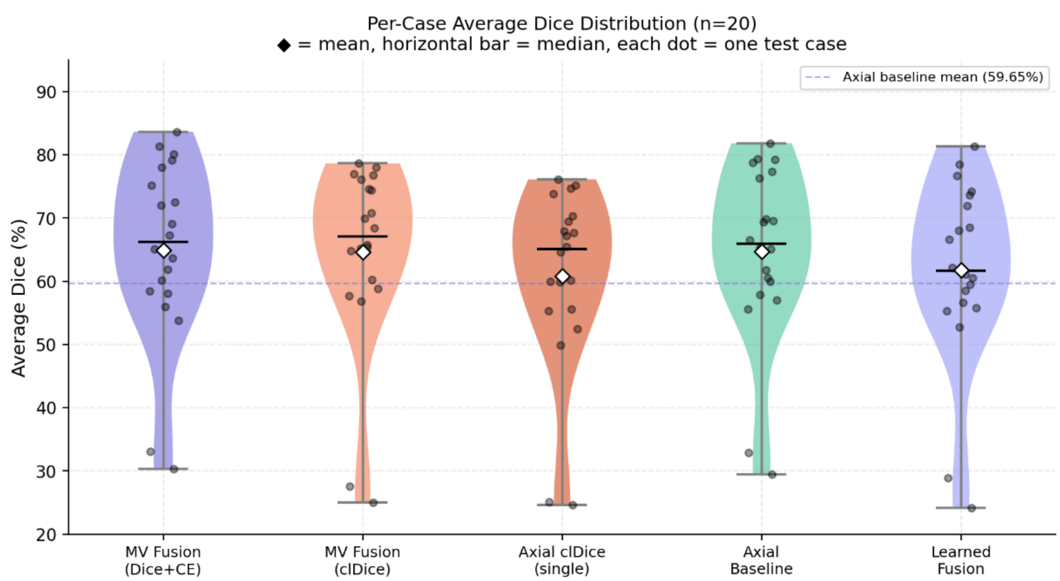

Figure 10. *Per-case average Dice distribution (n=20) for selected methods, shown as violin plots with individual case scores (dots) and mean (◆). The widespread confirms that case difficulty drives variance more than method choice.*

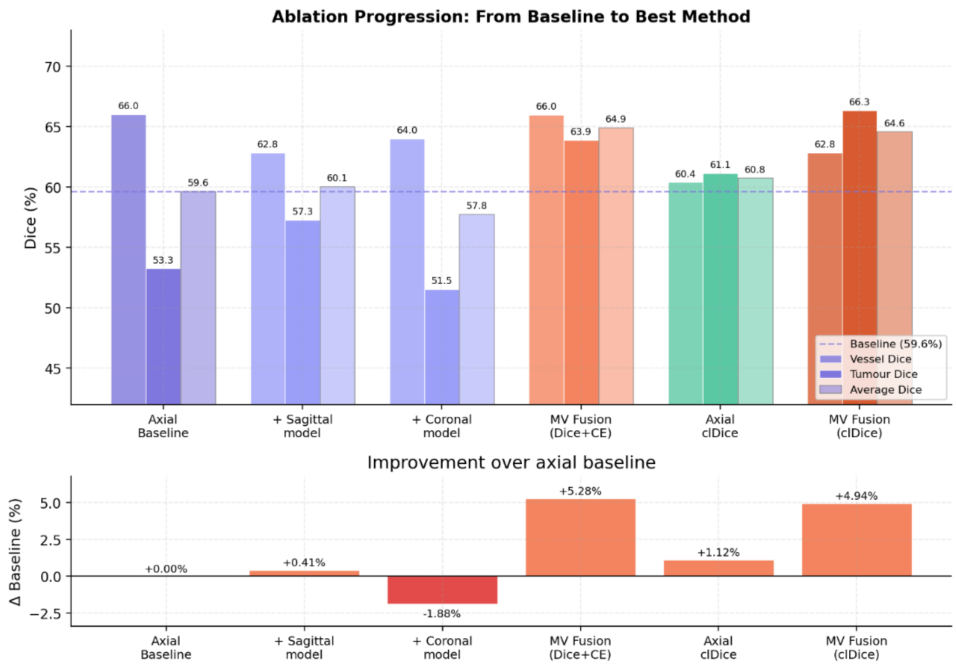

Figure 11. *Incremental ablation progression from axial baseline to best fusion method. Upper panel: Vessel (solid), Tumour (filled), and Average (faded) Dice per stage. Lower panel: Δ over baseline. Green bars = improvement; red = regression.*

Chapter 5. Discussion

5.1 Effectiveness of Multiplanar Fusion

The central finding of this project is that multiplanar majority voting fusion produces a statistically significant improvement over the single-orientation baseline. The +5.29% average Dice improvement (p = 0.0045) confirms our primary hypothesis (H1) that combining predictions from different anatomical orientations compensates for orientation-specific errors.

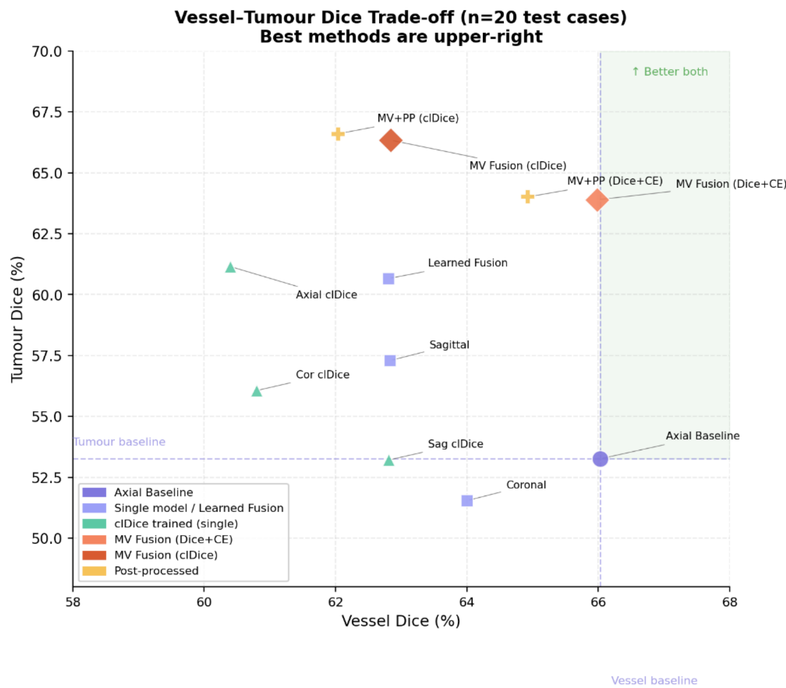

Figure 12. *Vessel–Tumour Dice trade-off across all methods (n=20 test cases). Methods in the upper-right quadrant achieve simultaneously high vessel and tumour performance. Dashed lines mark the axial baseline for each class. MV Fusion (clDice) achieves the best tumour Dice; MV Fusion (Dice+CE) the best overall balance.*

Figure 12 shows the vessel-tumour Dice trade-off across all methods. The upper right quadrant represents good performance on both classes simultaneously. Note the class asymmetry; Vessel Dice has increased slightly, not significantly, while tumour Dice has improved greatly. There was little change in Vessel Dice (65.98% vs 66.03%, p=0.81) and a large increase in tumour Dice (+10.63%, p=0.0029). A potential explanation can be seen from the sensitivity-precision trade-off. Majority voting is effectively a filter, decreasing the false positive rate while decreasing the sensitivity slightly. This is beneficial for tumours where the base model had very poor precision (44.88%). For vessels, which have good precision (66.45%) this effect becomes less relevant and is cancelled by the sensitivity decrease.

This observation matches the expectation raised by Perslev et al. [10], who hypothesized that the fusion of multiple views improves performance by reducing variance when the errors made by each view are uncorrelated. This is demonstrated by the three different orientation models that make unique errors - a vessel branch not resolved along the z axis in the axial model may be well resolved along z in the sagittal or coronal model - and majority voting that picks out voxels common to at least two models.

5.2 Why Majority Voting Outperforms Learned Fusion

As we predicted (H3), voting was better than learned fusion in our relatively small training set. The result is, however, an artifact that is important to consider. Our minimum learned fusion approach has an average Dice score of 61.73%, 3.20 percentage points lower than majority voting (64.93%). This is not surprising: the fusion CNN was constrained to fewer than 5,000 parameters and the network was trained using 64 patches from the 283 non-test cases. This was done to investigate whether it was possible to gain much from learned fusion given the low level of complexity that could be achieved here. This finding does not mean learned fusion is inferior to majority voting, just that at this level of complexity, it cannot be distinguished from majority voting.

This result is also consistent with the broader literature on ensemble learning.The work by Dang et al. [38] and follow-ups [45] suggest that learned ensemble weighting can be better than majority voting in high diversity, high training set size situations. In our project, neither case is applicable. The performance of the three orientation CNNs on the validation set is very similar (0.73/0.72/0.72), not much to play with in the weighting. Our CNN training set was much smaller than reported fusion CNNs were large. The fact that all three voting schemes give the same results, independently, seems to confirm this. If the performance is identical, any type of weighting, learned or not, cannot be expected to improve the performance substantially.

5.3 clDice Loss: Benefits and Instability

The hypothesis H2 is partially validated, and the nature of the partial validation is itself informative. Both clDice training (78.44%) and multiplanar fusion increase the topology sensitivity compared to the base axial model (71.61%), however clDice training does decrease the vessel Dice (60.40%) and precision, compared to the base model (66.03%). This is not an issue with clDice as the loss penalises false negative skeleton voxels which inherently causes the model to predict more towards recall rather than precision. It appears the clDice training and multiplanar fusion are actually tackling complementary issues of the segmentation task. Majority voting mostly increases the precision of the task (tumour precision is increased from 44.88% to 60.04%) because for a prediction to be classified as positive it has to agree across orientations. ClDice training mostly increases sensitivity of the task; it successfully recovered more skeleton voxels but had more false positives which degraded precision (vessel precision falls to 52.74%). The combination somewhat cancels the weaknesses out with MV Fusion (clDice) having a topology sensitivity higher than any individual method (75.65%), whilst still recovering much of the lost Dice from the single model clDice training (64.59% vs 60.77%).

Crucially, clDice training in isolation does not give any statistically significant increase in the average Dice score (p=0.9854) unless it is combined with multiplanar fusion to restore some of the lost precision that comes with its tendency towards recall. It is shown that choice between the two fusion methods should depend on the down-stream task; when boundary quality is important the use of MV Fusion (Dice+CE) would be preferred; when connectivity is important (and the ability to detect vessels is paramount) the use of MV Fusion (clDice) is preferable.

The instability observed at the higher weights for clDice (0.5 and 0.2) can likely be explained by conflict between the gradients produced by the clDice loss and the deep supervision utilized by nnU-Net. Deep supervision adds auxiliary losses to lower resolution, coarser levels of the decoder where it is not possible to have meaningful soft-skeletonisation (as the signal is much degraded) and the skeleton gradients may be confusing the main loss or conflicting with the high-resolution clDice signal. This type of failure was not present in the original clDice paper [20] which did not use deep supervision. Reducing the loss weight and increasing the warmup time is seen to eliminate the instability with a loss weight of 0.1, and simply running only one clDice loss on the final output resolution (ignoring auxiliary losses) remains a possible solution.

5.4 The Post-Processing Paradox

The expected benefits of connected-component post-processing were the elimination of small false positives, the increase of the Dice metric. It actually always worsened vessel Dice (1.05% for Dice+CE fusion, p = 0.0107; 0.80% for clDice fusion) with little to no gain on tumour Dice. The explanation for this effect is quite simple, at the selected spacing 1.5 x 0.799 x 0.799 mm, a 50-voxel component means an about 47.9mm³ segment; the diameter of a terminal peripheral vessel branch is on average 1 mm, and its length would be 5 mm at this resolution. The number of voxels corresponding to 1x5x1 mm with the same spacing would be approximately 40-60 voxels. So, it is obvious that this threshold is close to the genuine size of a peripheral branch. The threshold has not been calibrated on a validation set (which was not available), so the previous observation can be predicted, and it is reported more as a lesson learnt than an interesting result.

The only other interesting fact on the issue of connected component post-processing is that it affects cases differently; particularly the cases where anatomically more branching vessels are predicted or where the base Dice is lower. Those are the cases that should benefit more from an "improving" filtering; conversely, they get worse predicted because a larger percentage of genuine vessels fall below the threshold. This case-level variability implies that regardless of its value, a threshold in connected components is structurally inappropriate to HEPATIC VESSELSEGMENTATION. Learned reconnection like the one by Carneiro-Esteves et al. [42] or the use of the minimal component size in training data (minimum component size in the ground truth annotations) seems more suitable to solve this problem than threshold filtering on component size.

5.5 Comparison with Published Methods

Comparison to published benchmarks requires some nuance due to evaluation procedure differences: the benchmarks show a result on the actual MSD test set (140 cases), where we show a result on the 20 case validation set: The nnU-Net has 81.64% vessel Dice on the test set and our solution on the validation set had 66.03%. The difference is likely due to the larger evaluation set, possible differences in training (we only use fold 0, while the original submission of nnU-Net has a 5-fold ensemble), and standard error associated with a 20-case validation set. Despite this difference: our multi-planar fusion represents a 5.29% relative improvement over our own baseline. The improvement is statistically significant. We view our contribution as a novel training approach and not specific numbers on the test set. This would likely generalize to a nnU-Net with 5-fold ensembling fully trained.

5.6 Limitations

There are some caveats.

No submission to the official MSD test server. There are no publicly available labels for the 140 official MSD test cases - only the challenge server can be used to evaluate the results. This submission was not done by us, so we cannot compare our results to published benchmarks on a level playing field. Submitting at least the most successful method (MV Fusion Dice+CE) would have yielded a benchmark-comparable value and should be done as soon as possible.

Single fold training. We only train fold 0 for all our models, while the official nnU-Net uses 5-fold cross-validation and ensembling. This constrains our model's baseline performance and our absolute Dice scores are not as high as published 5-fold results.

Small evaluation set. The 20-case test set size is small and the large per-case variance (vessel Dice from ~47% to ~81%) means that individual cases can have a significant impact on the mean. Further, the 20 test cases were chosen alphabetically, rather than at random or stratified, and not evaluated for potential systematic bias.

Colab constraints. Time constraints, varying GPUs, and checkpoint-resume operations added constraints. The clDice training was especially interrupted and resumed several times, which could have impacted convergence.

No 5-fold ensembling of multiplanar models. A complete multiplanar pipeline would train all 5 folds of each of the 3 orientations (15 models), which was not possible due to the length of time required to train models (14 hours per model).

5.7 Why Not a Full 3D Volume Approach?

One central theoretical point to ask about this design is why we trained three different orientation-specific models, rather than taking the entire 3D volume, reorienting it in each direction, and training a single model on the entire volume each time. The reason is how the training volume size is dealt with in nnU-Net. Instead of processing the entire volume (even reoriented), nnU-Net uses patch-based training with a fixed patch size (64x192x192 for this problem) per volume. Every model has already seen 3D patches (as all training was 3D, not 2D with added channels for orientation), the orientation simply alters which anatomical axis is parallel to the x,y, or z direction in the patch volume. A patch of 64 voxels in z in the axial-trained network samples a different extent of anatomy than in the sagittal-trained network, for example where the z-axis has a different meaning. Training three different networks allows each to see different anatomical contexts from the same 3D volume through this differently ordered sampling of 3D patches.

As an experiment, we combined data from all three orientation-trained networks into a single Dataset (Dataset308) and trained a single model. The training failed completely and vessel Dice remained at 0.0%. The Dice score for vessels remained at 0% throughout the entire training procedure, suggesting the network could not, even with these three orientations combined into one training set, discern orientation-invariant vessel features.

5.8 Ethical Considerations

For this project, we employ the Medical Segmentation Decathlon Task08 dataset [40, 41] which is a completely anonymised benchmarkdataset available publicly with a license for open access for research applications. The CT volumes contained no patient identifying information and no ethics approval was sought. Computational tests use previously available data only and involved no patient interaction or clinical application. Although we envision these methods assisting clinical decision-making in the future, they are not validated for clinical applications and should not be used for patient care without subsequent approval and a prospective clinical trial. Computational experiments were conducted on cloud GPU (Google Colab Pro) with minimal environmental concerns above usual compute energy consumption.

Chapter 6. Conclusion

6.1 Summary of Findings

This thesis explores the use of a multiplanar deep learning method using the nnU-Net framework to enhance hepatic vessel segmentation on the Medical Segmentation Decathlon Task08 dataset, with all results reported on a 20-case held-out internal test split from the nnU-Net validation fold. Three 3D_fullres models were trained on axial, sagittal and coronal orientations of the data, and the results fused using majority voting. The main findings are:

1. **Multiplanar majority voting fusion improves segmentation.** Average Dice score has a statistically significant gain from the axial score (59.65%) to the fusion score (64.93%; $p = 0.0045$). The tumour Dice has the greatest improvement (+10.63%; $p = 0.0029$), mostly due to higher precision, from consensus filtering. It's difficult to say if this is clinically significant, as Dice acceptability thresholds for hepatic vessels have been reported widely, and are dependent on context. Nasser et al. [7] claims that branch topology (branch connectivity) is actually more important for surgical planning than Dice overlap, as branch junction disconnects in the segmentation could fool Couinaud segment calculations even if the Dice overlap is acceptable. This is where the MV Fusion (clDice) version (75.65% topology sensitivity compared to 68.64% for Dice+CE baseline) may be even more useful than Dice scoring. Further studies would be required to show this in a prospective study on patients, and ultimately the Dice scores obtained (64-65%) are still significantly lower than those obtained on the MSD test set with state-of-the-art methods, and far from clinical-research grade.

2. **Simple fusion outperforms a minimally learned fusion baseline under data-constrained conditions.** All three types of voting (majority, average, weighted) are identical since the quality of all models is very similar across orientations (0.73/0.72/0.72 val. Dice). A simple (very small CNN) learned fusion has a value of 3.20 percentage points below majority vote; this shows how good a learned fusion can be at this level of training data and not that learned fusions are necessarily worse.

3. **clDice training results in a precision-recall trade-off.** Axial clDice training increased sensitivity and tumour Dice, but decreased vessel Dice. ClDice-trained models combined with multiplanar fusion had the best topology (clDice = 70.47%) and tumour Dice (66.34%) scores, statistically equivalent to Dice+CE fusion overall.

4. **Post-processing hurts vessel segmentation**. Connected-component filtering removed valid peripheral vessels, hurting vessel Dice. This is a known negative outcome with implications for pipeline design.

5. **Combined-dataset training fails.** Pooling all orientations into one dataset resulted in zero vessel Dice, confirming that orientation-invariant vessel learning is not possible by pooling data. This confirms the need to model and then fuse.

6.2 Contributions

This project adds to the knowledge base of medical image segmentation as follows:

1. **Multiplanar fusion offers statistically and practically significant improvements for hepatic vessel segmentation in nnU-Net, via a precision-enhancement effect.** An average 5.29% Dice improvement (p=0.0045) is achieved, surprisingly, not through improved vessel sensitivity (actually slightly reduced), but through massive improvements in tumour precision (44.88 → 60.04%) which eliminates false positives through multi-orientation consensus. Existing multiplanar reports [10, 11] report a mixture of Dice improvement; but no work describes the underlying precision-recall improvement, nor the detailed per-class asymmetry; we provide these details in the current case study of CT hepatic vessels, where extremely high-class imbalance requires precision improvement to explain the Dice benefit 2.

2. **clDice loss training and multiplanar fusion have complementary effects on segmentation and give additive improvements.** As documented in Section 5.3, the topology-aware loss increases sensitivity, while multiplanar fusion increases precision; these are complementary effects which together result in the highest topology score (70.47%) and the highest tumour Dice among non-post-processed methods (66.34%); the post-processed variant achieves a marginally higher tumour Dice (66.60%) but is excluded as a best-method candidate given its statistically significant vessel Dice regression (−0.80%, p = 0.0107). Other works using topology-aware losses in ensemble methods [7, 37] did not breakdown sensitivity/precision, nor explicitly compare the effect of clDice loss and multiplanar fusion in nnU-Net.

3. **Integrating clDice loss into nnU-Net requires a conservative weight schedule to avoid training instability.** Weights of 0.5, 0.2 (as in the original clDice paper [20]) resulted in a diverging loss, and the weight had to be reduced to 0.1 and a warmup period of 50 epochs was required to converge. This instability was likely due to the competition between the topology gradient of clDice and nnU-Net's additional deep-supervised losses at earlier stages of the decoder, which was not mentioned in [20] since they did not use a deep-supervised design. This simple insight could be of immediate benefit to anyone looking to integrate topologically aware loss functions into a nnU-Net framework.

4**. Connected-component post-processing is contraindicated for hepatic vessel segmentation at standard size thresholds**. It worsened the vessel Dice by 1.05% (p=0.0107) due to the fact that true peripheral branches at 1.5 mm slices often have less than 50 voxels and cannot be separated from noise based on size; a quantitative explanation of the qualitative failure that Nasser et al. [7] reported.

6.3 Future Work

This project raises a number of questions that need to be answered beyond the implementation issues:

Does the precision-improvement mechanism of majority voting generalise across tubular structures? This project shows the improvement in the segmentation of hepatic vessels from multiplanar majority voting is due to a reduction of false positives, not to a large extent, the recovery of lost vessels. The mechanism may apply to coronary vessels, airways or retinal vessels; or it may only work on extreme class imbalance of hepatic CT. Regardless, the mechanism should be replicable on any structure on which the single model shows a large discrepancy of precision over sensitivity, as compared to sensitivity over precision; this discrepancy should be predictable from single model confusion matrices.

Is there a minimum model quality threshold below which majority voting degrades rather than improves? Our three models are all comparable quality (validation Dice 0.72-0.73), and all the improvements via fusion are significant. If one of the models was weak, e.g. Due to lower quality reoriented volumes or fewer training samples per orientation, then the errors of the worst model would start to be amplified by majority voting rather than reduced. By setting such a threshold, we would be able to make informed decisions about whether to include a certain orientation in the fusion or not.

Can soft probability fusion recover the topology benefits of clDice without the volumetric Dice cost? As we've seen, clDice fusion is more sensitive to topology, but less accurate (HD95) than Dice+CE fusion. Soft probability averaging before thresholding (this project did not have access to probabilities, so this experiment could not be run), may temper the precision-recall trade-off to produce a segmentation with both the boundary accuracy of Dice+CE and the connectivity sensitivity of clDice, the single best experiment.

Does the deep supervision interference hypothesis explain clDice instability, and can it be resolved architecturally? Restricting the clDice loss to the output of the full-resolution decoder while not including the deep supervision's auxiliary heads could test whether or not instability is a deep supervision problem. If the answer is yes, this may be a basis for a general recommendation on incorporation of topology-aware losses in any nnU-Net pipeline-not only for this hepatic vessel segmentation.

The next practical steps of implementation are: (1) 5-fold cross-validation with multiplanar ensembling (15 models, results much better than proposed architecture). (2) Submit to the offical MSD challenge server to obtain a benchmark evaluation result. (3) The Skeleton Recall Loss [23] can be used instead of clDice to avoid the GPU memory issue and, given a dataset with separate vascular labels (e.g., 3D-IRCADb-01), to enable multi-class segmentation separating portal and hepatic veins - a distinction that MSD Task08 does not provide, as label 1 combines all hepatic vessels.

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Appendix

A.

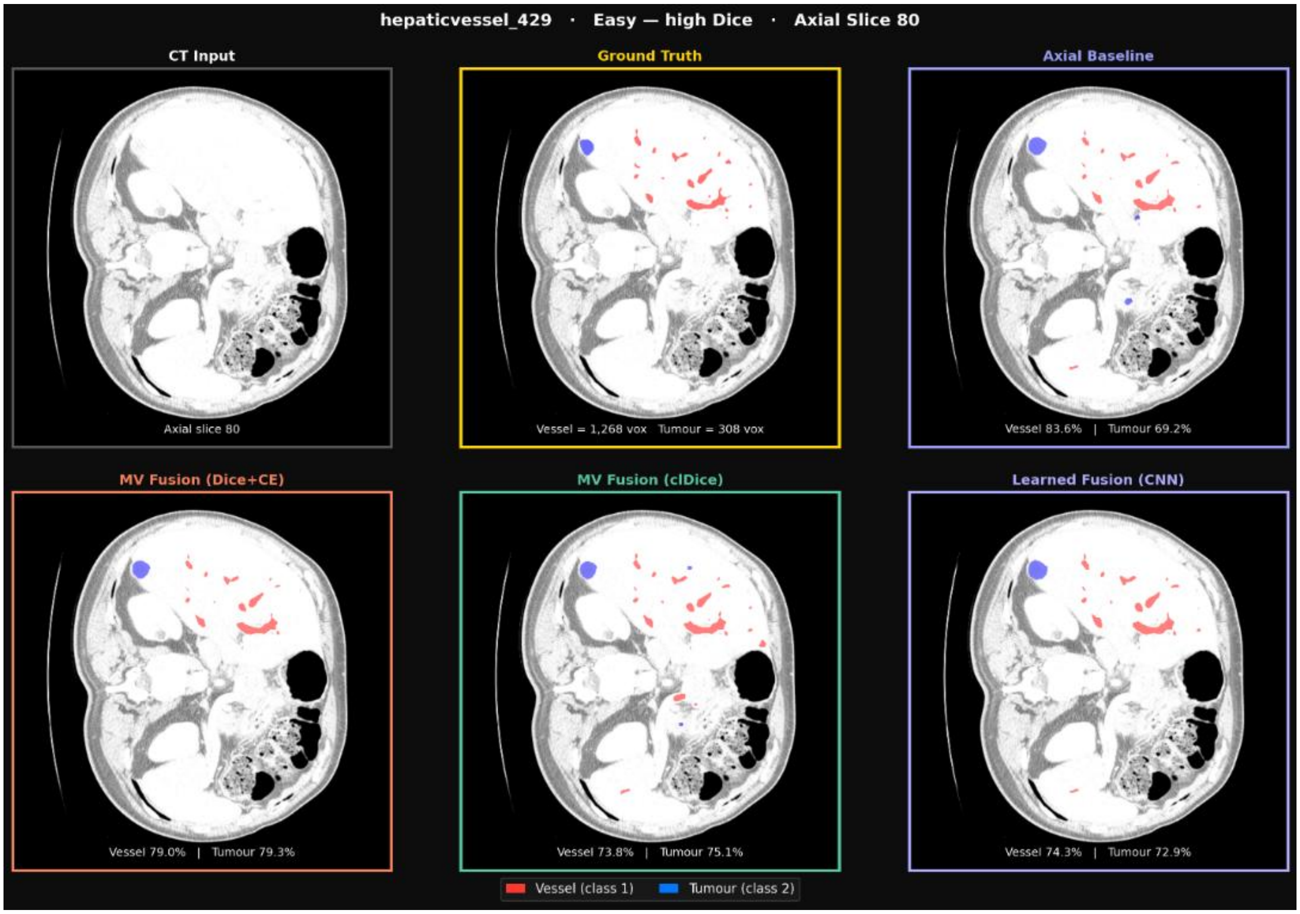

Figure A1. *Qualitative comparison on hepaticvessel_429 (easy case, high Dice), axial slice 80. Row 1: CT input, ground truth, axial baseline. Row 2: MV Fusion (Dice+CE), MV Fusion (clDice), Learned Fusion (CNN). Red = vessel, blue = tumour. Per-panel Dice scores computed on the 20-case held-out test set.*

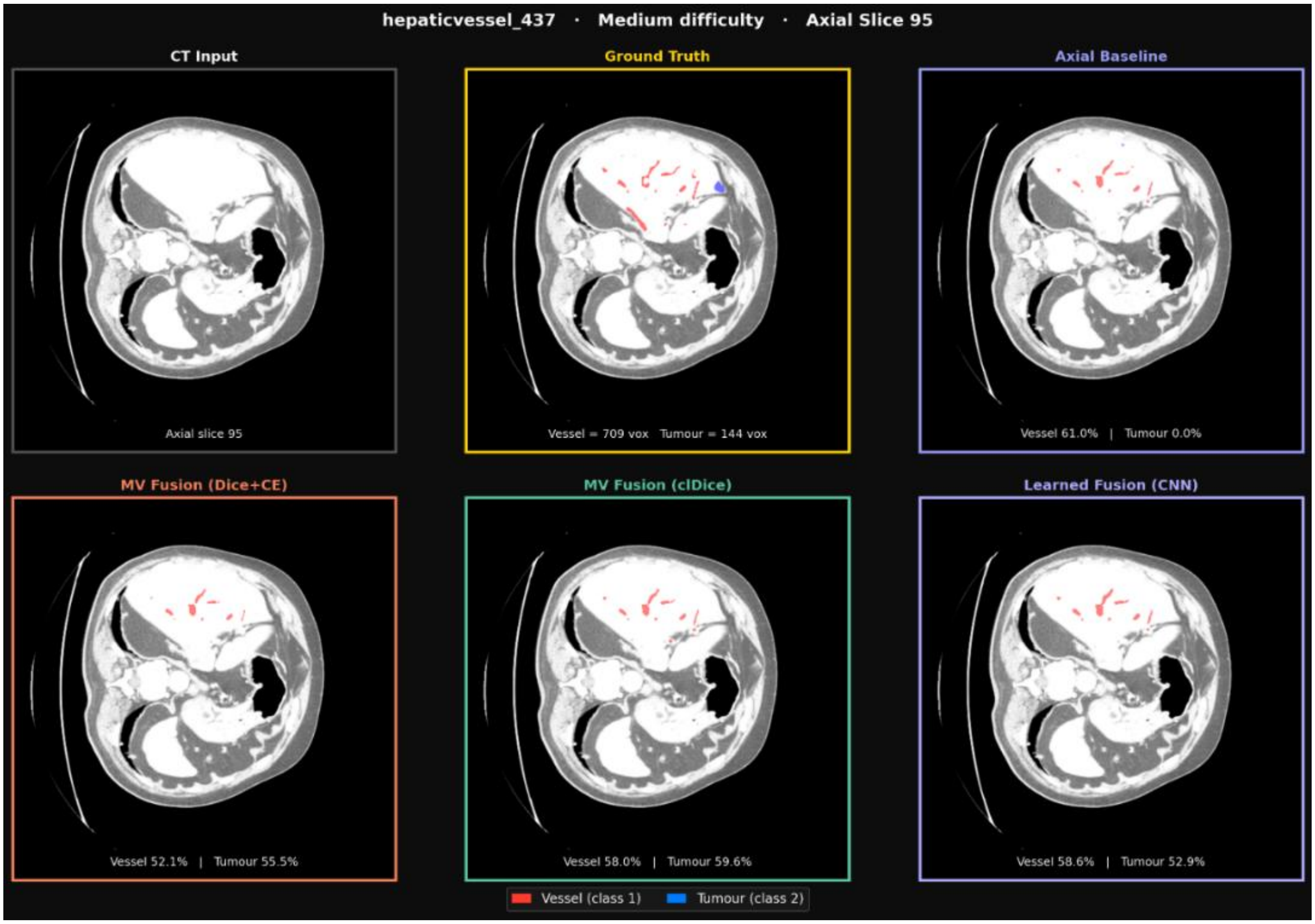

Figure A2. *Qualitative comparison on hepaticvessel_437 (medium difficulty), axial slice 95. Fusion methods show modest improvement over the axial baseline; the vessel tree is sparser and harder to delineate in this case.*

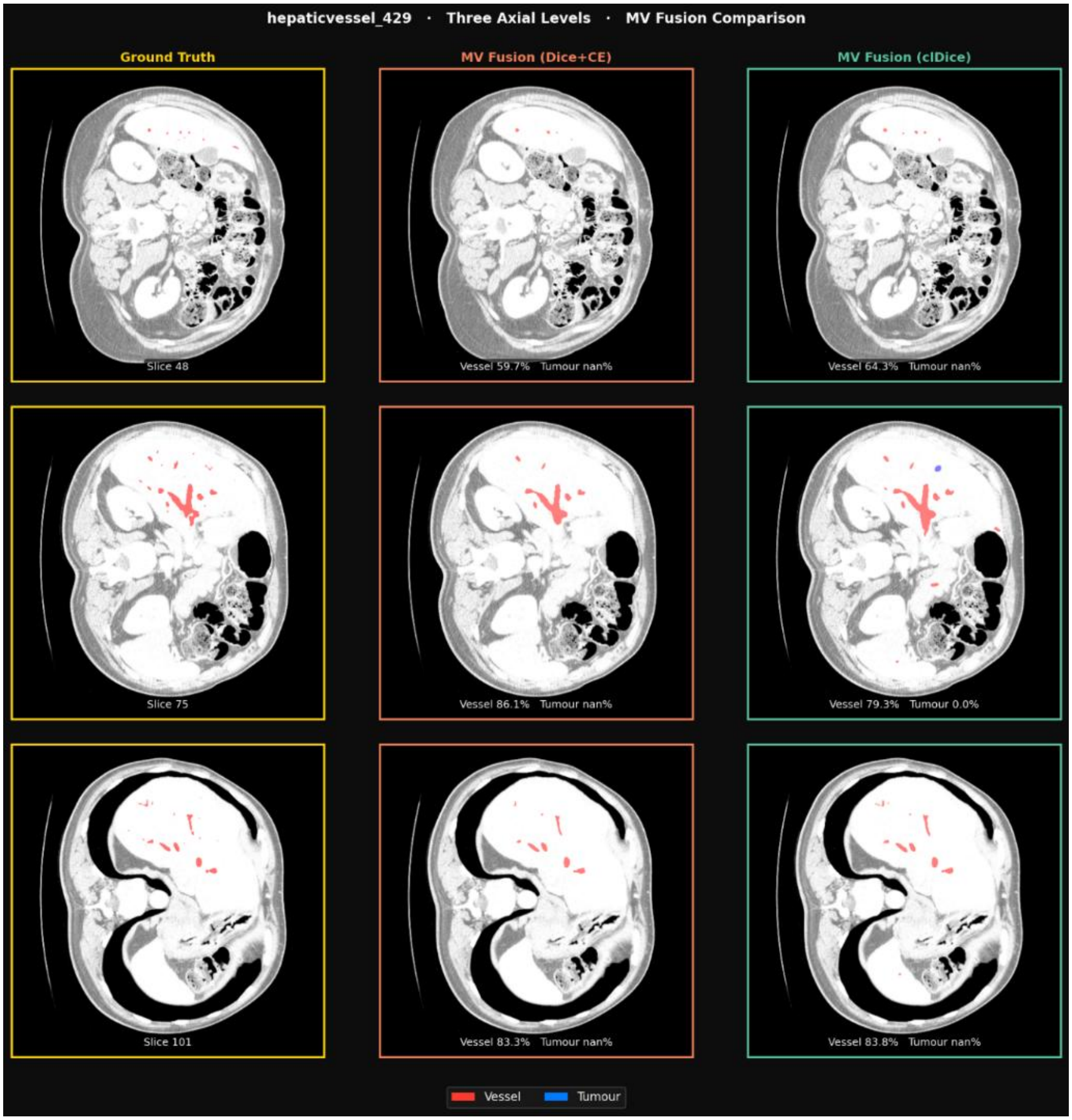

Figure A3. *Multi-level axial comparison on hepaticvessel_429 across three representative slices. MV Fusion (Dice+CE) and MV Fusion (clDice) are compared against ground truth at superior, middle, and inferior hepatic levels, demonstrating consistent segmentation quality across the full axial range.*

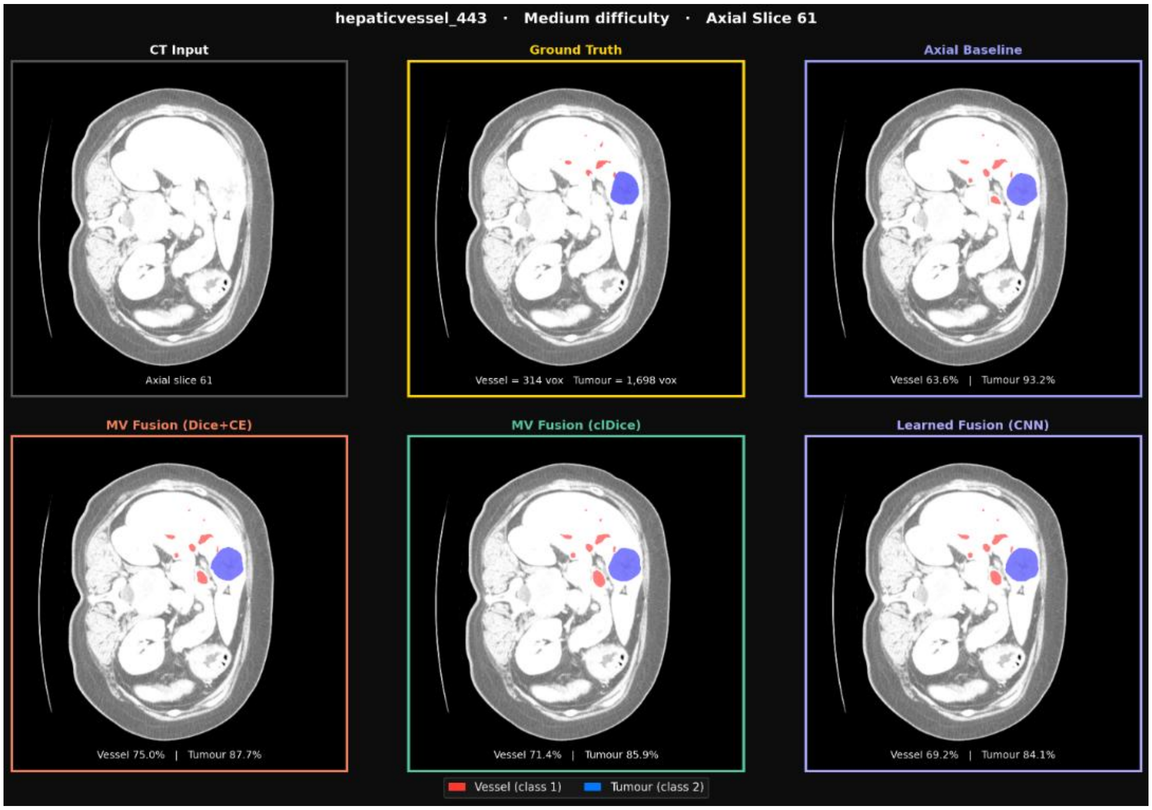

Figure A4. *Qualitative comparison on hepaticvessel_443 (medium difficulty), axial slice 61. All fusion methods preserve vessel connectivity relative to the axial baseline while improving tumour boundary delineation.*

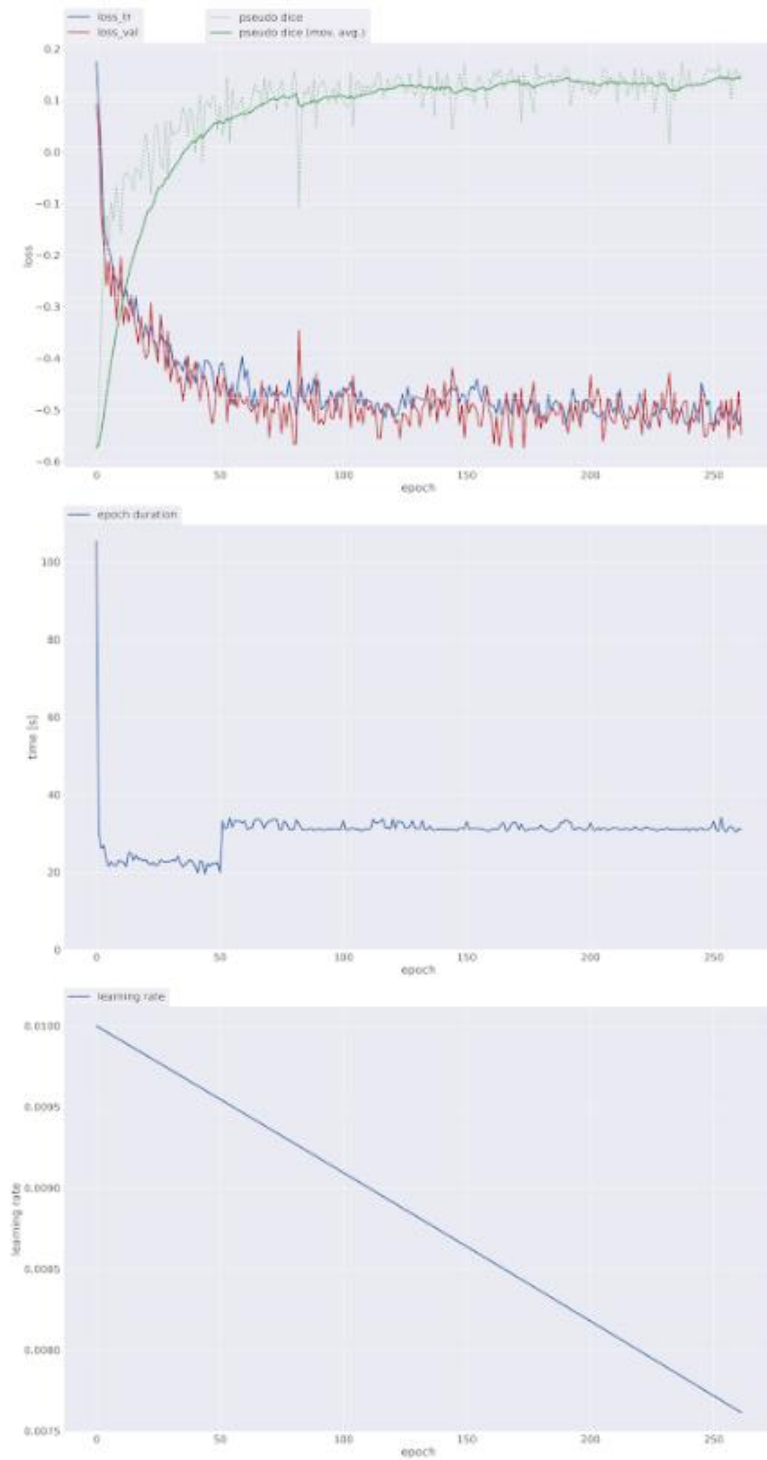

Figure A5. Training progression for MV Fusion (clDice) axial model. Top: training and validation loss with pseudo-Dice (moving average). Middle: epoch duration. Bottom: learning rate decay. The stable convergence after epoch 50 confirms that the warmup-then-ramp schedule (Section 3.5) successfully prevented the training instability observed at higher clDice weights.